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Artificial intelligence at work: A phenomenon-based interdisciplinary review and groundwork for multilevel scholarship

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ABSTRACT

The implications of artificial intelligence (AI) for work are significant and diverse, yet our understanding of its drivers remains siloed. This is partly due to a fragmented understanding of the AI phenomenon, its examination across diverse disciplines, and the contingent nature of its effects. We aim to help address these issues via two objectives. First, we *explore* the landscape of research by systematically reviewing how organizational science subdisciplines studying AI conceptualize, characterize, and investigate AI at work and then evaluate how this scholarship *clarifies* and *contextualizes* the phenomenon. By examining indicators of these dimensions, we identify distinct clusters of research that represent what we label as "application-orientation" and "generalized-orientation" categories. Comparatively, application-orientation research was the most likely to either define AI's capabilities concretely or to situate their assessments within a specific function or industry, was less likely to characterize AI as a radically or wholly new and disruptive technology, less likely to contain claims regarding widespread technological unemployment resulting from AI, and less likely to focus on the negative (compared to the positive) outcomes of AI use for workers. Comparatively, generalized-orientation research was less likely to reference AI's concrete capabilities or situate their analyses in a specific industry context, tended to be less empirical, and was more likely to position AI as radically disruptive or to focus on negative worker outcomes. Second, we seek to *add* to this research landscape by proposing an illustrative, interdisciplinary multilevel framework that suggests pathways toward balanced, multilevel assessments of the phenomenon.

1. Introduction

The development and use of artificial intelligence (AI) continues to surge, as evidenced by the $23 billion dollars privately invested in generative AI in 2023 alone [1]. Scholarly work in the field is also expanding rapidly (see [2]). Current AI technologies include machines achieving set goals with varied degrees of human supervision (via machine learning), machines deriving meaning from text and communicating in natural language (via natural language processing), machines discerning themes from large datasets (via topic content modeling; Copeland, [3]), and machines generating content, such as textual, visual, and audio output, through generative AI applications.

The rise of AI brings complex, competing, and significant implications for the future of work, including the reconfiguration of roles, shifts in human–machine work arrangements, and the need to re-align workers' skills [4]. However, beyond these broad implications, understanding AI's effects on different workplaces and the drivers of these effects warrants further specification in the organizational science literature [5]. Such assessments are complex, but we argue that they will particularly benefit from being grounded in *clarifying* the phenomenon through a clear articulation of AI and its capabilities as applied to particular purposes and *contextualizing* the phenomenon within the broader environment that influences AI's effects. Yet evidence suggests that diverse organizational science subdisciplines may differ in how

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fully and consistently they are engaging with these aspects, leading to varied narratives and approaches to the study of AI at work.

In terms of *clarity*, there can be varied, vague, or sometimes distorted characterizations of AI, leading to it remaining a "poorly understood organizational phenomenon" ([5], p. 408), with some arguing that "more precise vocabulary to describe the technical systems" is needed ([6], p. 10). Critics have pointed out that the popular press, academic literature, deploying organizations, and technology companies themselves can inflate and overstate the capabilities of current AI systems [7, 8], including through "AI washing" [9,10], which can lead to mischaracterizations and misunderstandings regarding its use in organizations [11–13]. This can also drive characterizations of AI as a wholly new-to-the-world phenomenon, obscuring its connections to past waves of technological change [14]. For example, most current forms of AI, while increasingly impressive, continue to rely on statistical pattern recognition, prediction, and particular analytic capabilities that can automate certain processes, meaning they share commonalities with prior waves of automation [12,15,16]. While each generation of technology brings new and advancing developments, established mechanisms and frameworks for examining how technology broadly transmits its effects on work exist in multiple fields, such as economics [17] and organization studies [18,19]. Such shared characteristics across waves of industrial revolutions include technology displacing workers, new jobs being created, potential productivity and output increases through changing roles [20], and ongoing implications for worker autonomy and control [18,21]. Yet, the "easy seductiveness" [22] of viewing emerging technologies as wholly new risks marginalizing the application of insights from prior seminal research on automation and computerization that remain highly relevant to studies of AI's effects on work today [23]. Ultimately, lack of clarity about AI's capabilities limits our understanding of the phenomenon itself, risks disconnecting it from lessons already learned through past technological disruptions, and impedes knowledge development regarding its current impacts on work.

Alongside clarity issues, an absence of *contextualization* can also limit organizational AI scholarship. The increasingly diverse range of AI technologies, use cases, and deployment settings means that, beyond clarity, it is important to also pay attention to a range of key factors—including organizational, institutional, and industry considerations—that shape AI's implementation. A lack of integrative scholarship across levels of analysis is a longstanding issue in the organizational science literature in general [24] and is an emerging constraint on its study of AI in particular [25]. Micro-level impacts of AI on work are contingent upon factors such as why the technology is adopted, how it is implemented, and how it interacts with the larger business and economic context. Such impacts will also vary based on the degree to which AI is used to substitute for current tasks, complement existing capabilities, or generate demand for entirely new tasks. The introduction of new technologies is thus heavily context-dependent [26], meaning that advancing a comprehensive assessment of AI at work requires accounting for the roles of meso- (i.e., organizational) and macro- (i.e., economic and institutional) level factors upon which micro-level employee effects are contingent [17,24].

A wider challenge to clarity and contextualization stems from the apparent fragmentation and siloed nature of research across diverse organizational science subdisciplines that study AI's workplace effects. These subdisciplines include economics, strategy, information systems, production and operations, marketing, tourism and hospitality, organizational psychology, critical management studies, and organizational behavior. This disconnection, particularly between the more technical subdisciplines that articulate, develop, and apply AI technologies to targeted problems and the social science subdisciplines that examine AI's implications, exacerbates clarity issues. Disciplinary norms and foci also lead to different vocabularies being used to study a phenomenon [27], a tendency toward focusing research on a single level of analysis, and prioritization of aspects of technology that align with disciplinary models [26]. This leads to a disjointed approach to studying a broad phenomenon such as AI and a fragmented discourse on its implications for work. A systematic review and integration of research across these subdisciplines is necessary and timely to help synthesize and develop a contextualized, coherent body of knowledge on AI's effects on work.

Overall, our aim is to provide an interdisciplinary critical evaluation of how diverse bodies of research are engaging with the phenomenon of AI and ultimately building narratives and insights around its nature, use, and implications for work. This evaluation is underpinned by a focus on spotlighting opportunities to more fully reintegrate into the AI discourse established models of technology's effects on work that have successfully conceptualized and evaluated those effects through earlier waves of technological change. We argue that more concerted efforts toward this end could assist in defusing some of the hype and fear regarding AI's impacts by connecting it to past technological disruptions and centering on balanced, phenomenon-led, and context-sensitive examinations of its effects. Our review makes two interconnected, but distinct, contributions.

In Part 1, we seek to *explore* and *understand* the landscape of research by systematically reviewing and then categorizing how diverse subdisciplines in the organizational science literature conceptualize, characterize, and investigate AI at work. From this foundation, we evaluate how this scholarship *clarifies* and *contextualizes* the phenomenon. Using bibliometric techniques to map the intellectual structure of the literature [28], we first identify distinct clusters of research based on their shared citation patterns. We then manually code a subsample of articles to evaluate their characterizations of AI, the extent of contextualized multilevel analyses, and their engagement with extant models of technological change, particularly the task-based model of substitution–complementarity–new task generation [17]. While we acknowledge that this model derives from the economics subdiscipline, its three channels also helpfully capture characterizations of AI's impacts that other disciplinary work may more broadly discuss as worker displacement/replacement (substitution) and human–AI collaboration (complementarity).

In Part 2, building out from the limitations identified through the Part 1 review and evaluation, we then seek to *add* to the existing research landscape by proposing an illustrative, interdisciplinary multilevel framework that suggests a pathway to link the study of AI's effects on workers' jobs (at the micro level) to its contexts of use by identifying salient factors at the meso (i.e., managerial choices associated with technological transformations) and macro (i.e., economic forces) levels of analysis. In doing so, we draw upon prior models of the impacts of automation and computerization on the workplace from across disciplines, to acknowledge the continuity of AI with past technologies while still accounting for its unique features. Our goal is to leverage the value that diverse disciplinary perspectives can collectively contribute to this endeavor. The spine of our multilevel framework remains linked to the substitution–complementarity–new task generation framework, as we articulate how particular multilevel factors influence whether AI's effects are more or less focused on each of these channels. This framework offers possible pathways for future research to better illuminate the complex and contingent nature of AI's varied impacts on work.

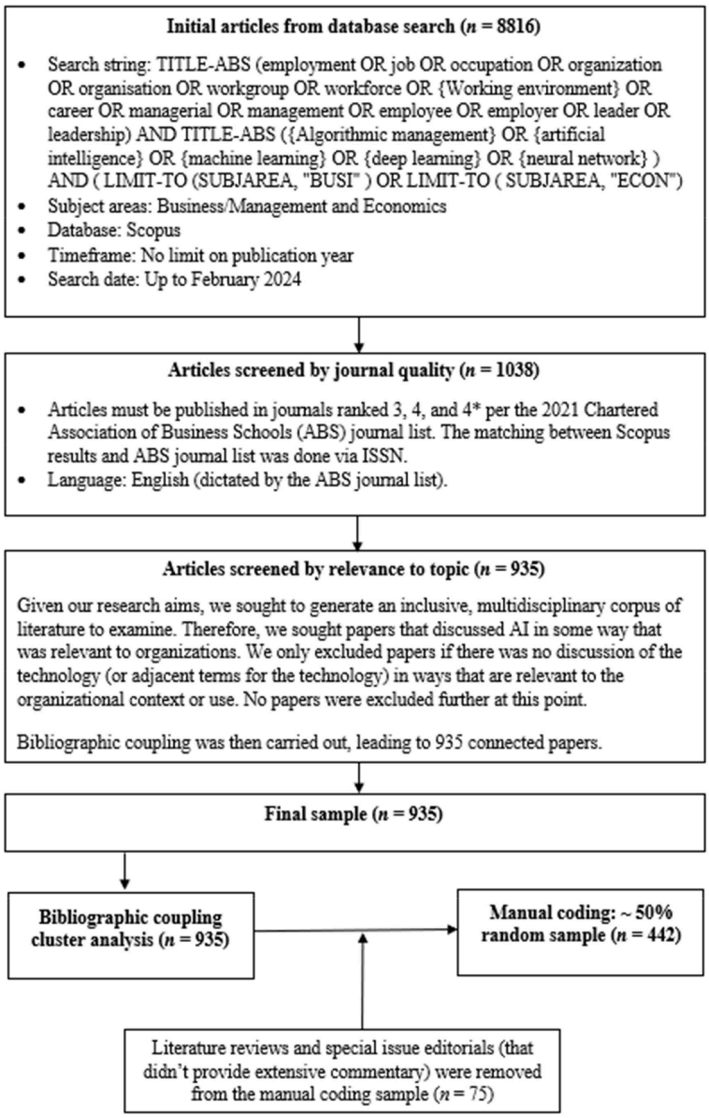

Fig. 1. Flow diagram of search process and inclusion criteria.

2. Part 1: systematic review and critical assessment of the literature

2.1. Article search process and analytical approach

Our search process is visualized in Fig. 1. To achieve our aims, we sought to generate an inclusive and diverse corpus of literature representing multiple disciplines. This is because the study of AI, even within organizational contexts, is a broad and diverse disciplinary endeavor. While systematic literature reviews can follow narrower paths, extracting the value from taking a wider view [29] and generating breadth was a core focus of our review and guided our search terms and inclusion/exclusion criteria. Since we aimed to identify research that in some way examined AI in organizations, our search terms encompassed "organization" or "employment" or "workforce" or "management" (and adjacent terms) that would capture some form of organizational AI use, alongside terms that reflect AI, including other common terminology such as "neural network" and "machine learning". The full search string is in Fig. 1 (Box 1).

We employed a journal-driven approach [30] to identify relevant research, drawing from the 2021 *Academic Journal Guide* (provided by the Chartered Association of Business Schools). We did not impose further methodological or theoretical exclusion criteria, aligning with our aim of generating a discipline-diverse literature sample. This is because disciplinary norms regarding methods and theory can vary widely, meaning we deliberately avoided applying exclusion criteria based on our own team's disciplinary conventions to ensure an inclusive sample. Overall, our approach was intentionally inclusive, incorporating papers that discussed AI in some way that was relevant to an organizational context or use.

We then conducted two complementary analyses. First, we applied bibliometric analyses, via bibliographic coupling, to the sample of

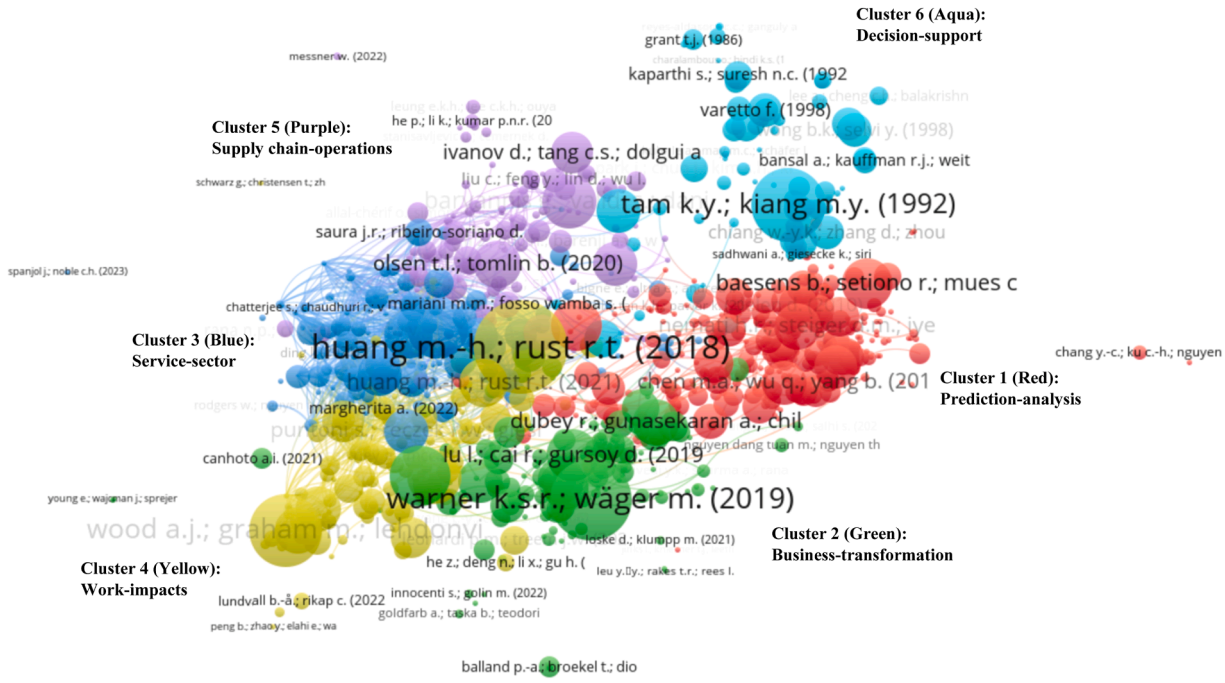

Fig. 2. Visualization of the six research clusters based on bibliographic coupling ($n = 935$).

papers initially screened by journal quality ($n = 1038$). This analysis revealed a network of 935 connected papers in VosViewer. The bibliographic coupling, which generated the clusters of research, was performed on this sample of connected papers ($n = 935$).[2] This method allowed us to establish the current research front of organizational science literature on AI at work by generating clusters within this literature that focused on similar subject matter ([28]; [31]).[3]

In the bibliographic coupling methodology, papers that share a significant number of common references are more likely to be grouped together, as shared citation patterns suggest that they address related topics or similar research questions. We selected bibliographic coupling for its accuracy in grouping research by intellectual lineage and research objectives. To generate the clusters, we used VosViewer 1.6.5 software with a resolution of 0.60, accounting for the large number of papers and ensuring the formation of meaningful and coherent clusters, while maintaining a minimum cluster size of 25 research items.

Second, we generated a manageable scope of papers for more labor-intensive analysis. To do this, from our sample of connected papers ($n = 935$), we manually coded a randomly selected subsample of 50 % of papers from each cluster ($n = 442$ papers in total) to identify indicators associated with the aspects of *clarification* and *contextualization*. Given the scale of our literature sample, this approach allowed us to balance a robust evaluation of this literature with the practical demands of extensive manual coding. As noted in Fig. 1, because we adopted a systematic review approach, we did not manually code other literature review papers or special issue editorials that did not provide extensive discussion of AI (i.e., they only provided an overview of the special issue papers); however, we did include these papers in the bibliometric analyses ($n = 935$). Our manual coding scheme[4] focused on general descriptors of each article (e.g., empirical/theoretical; methods used) and indicators relating to the clarity and contextualization of AI through, for example: definitional and functional characterizations of AI; broader characterizations of AI; connection to past work on automation; AI's impact on workers; and focal level[s] of analysis.

The manual coding was conducted by a team of 14 coders: three members of the author team and 11 research assistants (RAs). The RAs, recruited as postgraduate students in advanced coursework, were already familiar with reading and interpreting scholarly articles. Coder training involved a first round where all coders, including the author team, were sent three sample papers to code using the coding scheme. During the online training session, the first author explained each aspect of the coding scheme and demonstrated how to apply it to the example papers. Following this, each RA was provided with the same 10 papers to pilot code. These results were then used to assess consistency of coding by checking for coder fixed effects on our key variables. In almost all cases, the effects were not significantly different from zero, with only a handful of significant deviations. Only one variable (i.e., an indicator for meso-level analysis) showed a pattern of multiple significant effects. We note that all conclusions drawn in this review are robust to re-assigning a substantial portion of the meso papers to micro or macro status. Before distributing the full set of papers for manual coding to the RAs, the first author provided them with individual feedback on their pilot coding, to resolve any uncertainties or discrepancies. Unless otherwise stated, our findings are based on statistical analyses of the manually coded papers using Stata statistical software.

We now present the findings from these two sets of analyses by outlining the intellectual structure of the corpus (via the bibliographic coupling) and then presenting our review of it across the focal areas of *clarifying* and *contextualizing* AI at work (via analyzing the manually coded sample of papers). We conclude the section by critically assessing this literature.

[2] Details of the full sample of papers are available at https://figshare.com/s/802e2ab7e122e6ba1d0c.

[3] For readers with an interest in the evolution of these clusters, we also conducted a quantile-based analysis to establish meaningful temporal segments within each identified cluster, to track the evolution of their research themes systematically. This additional, supplemental analysis is available at http://doi.org/10.25949/30060103 (and is not presented or discussed further here).

[4] The base coding sheets used are available at https://www.doi.org/10.25949/30059095.

2.2. Intellectual structure of the literature: bibliographic coupling ($n = 935$)

2.2.1. Individual cluster naming approach

Fig. 2 visualizes the six clusters generated.[5] It is important to reiterate that the clusters were generated via bibliographic coupling analysis and, while the clusters are not mutually exclusive, the papers in a particular cluster are grouped based on their shared citation patterns. The processes we undertook to name and describe these clusters focused on their most salient features. To do this, we undertook a two-step approach using both machine and human intelligence: (1) we calculated the term frequency for words present in the abstract and title of each publication, for each cluster; and (2) the first author scanned the titles and abstracts of all papers in each cluster and categorized them thematically, based on the features and focus of each paper. The results of these steps, and the labels provided to each cluster, are presented in Table 1. Table 2 provides a descriptive overview of the clusters.[6] While we strive for clarity and distinctiveness in our cluster names, some themes may overlap across clusters, and this reflects the complex and diverse nature of AI research.

Overall, our cluster naming and descriptive conventions are intended to be heuristic tools to aid in understanding, navigating, and evaluating this large and diverse body of literature. These conventions are not intended to provide definitive, inflexible demarcations and are somewhat porous. However, such heuristics allow us to highlight distinct research streams and the interconnections between different areas of inquiry, and ultimately to offer a broader categorization of the clusters based on our indicators of *clarification* and *contextualization*.

2.2.2. Cluster naming and descriptions

Cluster 1 (Prediction-analysis cluster): AI for analyzing complex problems. Papers in this cluster examine how AI can be applied to a diverse range of complex predictive and analytical tasks, across a range of sectors, that depend on the integration of multiple and possibly unstructured sources of data for pattern detection to solve problems. For example, papers in this cluster examine applications of AI to areas such as fraud detection [32], water supply planning [33], and forecasting new product demand [34].

Cluster 2 (Business-transformation): AI and business performance. This cluster broadly centers on how AI generates business value and enhances firms' performance, across varied sectors, through innovation-led processes [37]. It includes papers with a technical focus that explicitly develop or apply AI technologies to help solve business problems (e.g., [36,50]). Other papers examine the factors influencing organizational technology adoption [35] and how AI can help address societal problems, such as climate change [51]. Papers in this cluster also canvas a wider range of issues that emerge from changing business processes and that are evident in other clusters, such as ethical implications, impacts on the nature of work, and human–AI collaboration.

Cluster 3 (Service-sector cluster): AI in services. This cluster examines AI use in service sectors, such as retail, tourism, and marketing. It focuses on the use of AI, such as through conversational agents like chatbots [39], for consumer engagement and experience [40] and customer relationship management [52], and the influence on service quality [53]. Impacts upon the marketing function are also discussed [38], alongside implications for workers in service sectors [54]. When AI is discussed as an analytical tool, it is usually applied to examining social media use or other text-based data for marketing insights [55].

Cluster 4 (Work-impacts cluster): AI's work implications. This cluster most clearly focuses on AI's implications for workers and the wider experience of work. It particularly centers on AI's influence on the nature of human work [56] and emerging forms of human–AI collaboration [41], while also canvassing the ethical implications of these changes [42]. These articles also assess the effects of algorithmic management for platform workers and those in gig work arrangements [43].

Cluster 5 (Supply chain-operations cluster): AI in supply chain and operations management. This cluster discusses how AI is used to support manufacturing and supply chain and operations management. The types of problems that AI is applied to here relate to optimizations, such as in job shop scheduling [57] and in manufacturing [45]. Other papers highlight the implications of such technologies in relevant sectors, such as for enhancing supply chain resilience [46] and purchasing functions [44].

Cluster 6 (Decision-support cluster): AI analytics for decision making. A key focus here is examining how AI can be used to extract insights from large volumes of data generated within organizations or across sectors, by leveraging its predictive capabilities and discussing how AI technologies can support organizational decision making in some form (e.g., [48,49,58]). Compared with other clusters, it offers a more historical perspective on AI (age range of papers: 1986–2023; average age of publications: 2006). Many papers in this cluster position AI capabilities as enhancements of past technologies, for example, by contrasting neural network predictive capabilities with other forms of modeling (e.g., [47,59,60]).

We now critically review this literature across our *clarity* and *contextualization* indicators.

2.3. Reviewing the literature for clarifying and contextualizing AI: manual coding ($n = 442$)

2.3.1. Reviewing clusters regarding clarity of the phenomenon

Definitions and functional characterizations of AI. Few papers in any clusters provided formal definitions of AI. Overall, only 16 % of articles contained an explicit definition. The work-impacts and service-sector clusters had the highest percentages of papers with AI definitions, with 26 % and 40 %, respectively. A sample definition of AI from the work-impacts cluster is as follows: "Artificial intelligence (AI) refers to computer systems that perform tasks that normally require human intelligence, such as visual perception, speech recognition, decision making or translation" ([61], p. 1058). In the prediction-analysis, supply chain-operations, and decision-support clusters, between 3 % and 15 % of papers defined the term. The business-transformation cluster had an intermediate 20 % level on this measure.

Because research focused on the development of AI for specific purposes may instead discuss or define specific AI technical capabilities, without necessarily defining the overarching phenomenon, we also examined whether papers provided such discussions. There was a wide variation among clusters in the discussion of AI's specific technical capabilities. The decision-support (89 %) and supply chain-operations (81 %) clusters had the highest percentages of such papers, followed by the prediction-analysis cluster (76 %). The work-impacts cluster had the lowest percentage of papers discussing AI's technical capabilities (53 %). A sample statement of AI's technical capabilities from the prediction-analysis cluster is the following: "ANNs (artificial neural networks) can be used to derive a wide variety of non-linear mathematical models, applicable to many types of statistical problems" ([62], p. 10).

We also created a measure of whether a given paper included *either*

[5] The bibliographic coupling analysis initially identified seven clusters of research. The seventh cluster was by far the smallest ($n = 41$ papers) and during our initial review and naming of the clusters we observed substantial thematic and conceptual overlap between clusters 7 and 1. Specifically, the research in cluster 7 shared methodologies and topical focus with cluster 1, making it difficult to justify their separation as distinct clusters. Given these similarities and the small size of cluster 7, we deemed it appropriate to merge the two clusters to avoid redundancy and to provide a clearer, more coherent grouping of the research. This resulted in our final six distinct clusters of research.

[6] For further descriptive information on our sample, Appendix 1 overviews the distribution of publications in our sample across journals and Appendix 2 identifies the top five high-impact publications (by annual citation rate) in each cluster.

Table 1
Term frequency (machine) and thematic (human) analysis for cluster naming (n = 935).

Cluster number	Most common keywords (term frequency analysis)	Themes discerned from title/abstract scan (author analysis): the top five themes for each cluster	Final cluster name
Cluster 1	management: 142 system: 138 decision: 125 network: 123 analysis: 101 neural: 90 data: 81 forecasting: 72 model: 70 deep: 69	Prediction Forecasting AI applied to analyze data (diverse uses) Classification Detection	Prediction-analysis
Cluster 2	technology: 104 management: 81 data: 73 innovation: 71 analysis: 57 decision: 54 research: 52 system: 45 making: 45 development: 40	AI applied to analyze data (diverse uses) AI adoption Organizational performance Innovation Occupational-specific insights; addressing social problems	Business-transformation
Cluster 3	management: 26 decision: 12 information: 9 technology: 8 theory: 8 system: 7 social: 7 human: 7 service: 7 making: 7	Consumers/customer relationship management AI applied to analyze data (diverse uses) Marketing Worker impacts (often in service/hospitality sectors) Chatbots/conversational agents	Service-sector
Cluster 4	management: 24 system: 20 labor: 18 automation: 17 human: 16 technological: 14 research: 14 employment: 14 technology: 14 information: 13	Nature of work Algorithmic management /gig work Human-AI collaboration Fairness/ethics Occupational-specific insights	Work-impacts
Cluster 5	management: 78 supply: 71 chain: 67 system: 56 data: 46 manufacturing: 35 network: 32 decision: 29 research: 25 technology: 24	Supply chain Operations Manufacturing Production Scheduling	Supply chain-operations
Cluster 6	system: 49 network: 32 decision: 30 neural: 29 control: 23 management: 19 data: 18 analysis: 18 scheduling: 17 process: 16	Decision support Scheduling AI applied to analyze data (diverse uses) Prediction Forecasting; neural network theory; risk modeling	Decision-support

an AI definition or a discussion of AI's technical capabilities to create an overall assessment of definitional and functional clarity within our sample. The decision-support and supply chain-operations clusters had the highest percentages of papers on this combined variable (85 % and 93 %), followed by the prediction-analysis and service-sector clusters (77 % and 79 %), while the work-impacts and business-transformation clusters had the lowest proportions (across all clusters) for this combined variable (63 % and 67 %). This pattern of results suggests that over a third of papers focused on the impact of AI on work don't concretely define AI nor discuss its technical capabilities in some way, while more technically oriented and applied clusters had the highest proportions of papers discussing AI's capabilities.

General characterizations of AI. Beyond definitional and functional discussions, broader characterizations of the nature of AI can inform clarity discussions, given that these can span from revolutionary to more circumspect [63,64], to characterizing generative AI as "the most important of the technology revolutions" ([65], referencing the CEO of OpenAI, Sam Altman) or viewing such tools as being "still just a prediction machine" [66]. Uncovering these characterizations goes some way toward understanding the narratives surrounding AI that different disciplines focus upon and perpetuate [7], which can then frame how researchers study the phenomenon. We assessed such characterizations by asking "Does the article identify AI as a disruptive/completely new/revolutionary technology?" and identifying whether papers discussed AI as "disruptive" or as a "wholly new" or "radically different" technology.[7] The business-transformation and work-impacts clusters had the highest percentages of papers characterizing AI in this way (42 % each). For example, within the work-impacts cluster, Last [[67], p. 48] notes that "the consequences of an 'exponential' and 'global' technological revolution are almost or even totally unpredictable in the sense that the structure of human life and civilization will undergo changes of a unique qualitative nature." Conversely, the prediction-analysis (8 %) and decision-support (0 %) clusters had the fewest papers discussing AI in these ways, and about a quarter of supply chain-operations and service-sector papers did so (23 % and 30 %, respectively). These findings indicate that a large proportion of the research specifically focused on how AI will impact work characterizes it as a radically new and disruptive technology, in contrast to more technically oriented research and studies examining AI's application in specific contexts.

Incorporation of prior work on automation/technological processes. We also assessed whether research in our sample acknowledges and builds upon the insights of relevant prior research to investigate the effects of AI. We did this in two, and we acknowledge bounded, ways. First, we draw back to the widely cited task-based model of the impact of technology on work, with its tenets also more broadly applied in other disciplines, which centers on the established channels of technology *substituting* for workers (i.e., AI replacing humans), *complementing* workers (i.e., AI stimulating demand for certain human skills and supporting/augmenting human workers), or *generating new tasks* for workers (i.e., using AI creates new tasks for humans; [68]). We examined whether papers in our clusters utilized this model and its terminology in some way (without necessarily specifically referencing the "task-based model").[8] Most clusters did not show evidence of grounding their analyses in the processes of substitution, complementarity, and new task generation. Substitution and complementarity were the most widely referenced processes across all clusters, but only 22 % and 28 % of papers referenced each process, respectively. New task generation was mentioned by only 9 % of papers overall. Within clusters, the work-impacts, service-sector, and business-transformation clusters had the highest proportions of papers mentioning substitution (47 %, 40 %, and 34 %, respectively) and complementarity (50 %, 46 %, and 38 % respectively). New task generation was most commonly mentioned in

[7] This coding question asked "Does the article identify AI as a disruptive/completely new/revolutionary technology?" (Yes, No, Unclear). Within each paper the coding team were then guided to search for the following terms: disruptive/disruption/disrupt; radical/radically change; profound change; wholly new/different technology; highly distinctive.

[8] These coding questions asked, "Does the paper discuss substitute/substitution?", "Does the paper discuss complement/complementarity/human–AI collaboration?", "Does the paper discuss new task generation/new human work?" (Yes, No, Unclear).

Table 2
Cluster descriptives for full paper sample ($n = 935$).

Cluster	Cluster label	Description	Age of publications	Example journals	Example papers
Cluster 1: *AI for analyzing complex problems* $n = 347$ % manually coded: 53 % ($n = 173$; from a total of 328 non-literature review papers in the cluster)	Prediction-analysis cluster (application-orientation cluster)	Focus is on applying AI to a diversity of complex predictive and analytical tasks, across a range of sectors, that depend on the integration of multiple and possibly unstructured sources of data for pattern detection to solve problems.	Range: 1988–2023 Avg: 2019	*Decision Support Systems; International Journal of Production Economics; Management Science; Manufacturing and Service Operations Management*	[32–34]
Cluster 2: *AI and business performance* $n = 144$ % manually coded: 51 % ($n = 64$; from a total of 125 non-literature review papers in the cluster)	Business-transformation cluster (generalized-orientation cluster)	Focus is on how AI generates business value and enhances firms' performance, across varied sectors, through innovation-led processes. It also canvasses ethical and work-related issues associated with changing business processes.	Range: 1992–2023 Avg: 2020	*International Journal of Production Economics; Technological Forecasting and Social Change; Technovation; Research Policy*	[35–37]
Cluster 3: *AI in services* $n = 135$ % manually coded: 55 % ($n = 67$; from a total of 121 non-literature review papers in the cluster)	Service-sector cluster (application-orientation cluster)	Focus is on AI use in service sectors, such as retail, tourism, and marketing. For example, for consumer engagement, customer relationship management, and service quality purposes.	Range: 2007–2023 Avg: 2021	*Industrial Marketing Management; International Journal of Contemporary Hospitality Management; Journal of Business Research; Journal of Service Research*	[38–40]
Cluster 4: AI's work and worker implications $n = 130$ % manually coded: 51 % ($n = 62$; from a total of 122 non-literature review papers in the cluster)	Work-impacts cluster (generalized-orientation cluster)	Focus is on AI's influence on the nature of human work, emerging forms of human-AI collaboration, algorithmic management, and the ethical implications of these changes.	Range: 1999–2023 Avg: 2021	*Human Resource Management Review; International Journal of Human Resource Management; Journal of Business Ethics; New Technology, Work and Employment;*	[41–43]
Cluster 5: *AI in supply chain and operations* $n = 108$ % manually coded: 48 % ($n = 47$; from a total of 98 non-literature review papers in the cluster)	Supply chain-operations cluster (application-orientation cluster)	Focus is on how AI is used to support supply chain, manufacturing, and operations management.	Range: 2003–2023 Avg: 2021	*International Journal of Production Research; Journal of Business Research; Production and Operations Management; Production Planning and Control*	[44–46,]
Cluster 6: *AI analytics for decision making* $n = 71$ % manually coded: 44 % ($n = 29$; from a total of 66 non-literature review papers in the cluster)	Decision-support cluster (application-orientation cluster)	Focus is on how AI can be used to extract insights from large volumes of data generated within organizations or across sectors, by leveraging its predictive capabilities and discussing how AI technologies can support organizational decision making in some form.	Range: 1986–2023 Avg: 2006	*Information and Management; International Journal of Production Research; Journal of the Operational Research Society; Production Planning and Control*	[47–49]

the work-impacts cluster, but only by 24 % of its papers. The prediction-analysis, supply chain-operations, and decision-support clusters had some of the lowest proportions discussing any of the three channels. This finding indicates that research across all clusters, even when focused on AI's work implications, is often not taking the established, full heterogeneity of technological interactions with human labor into account.

Second, we examined whether researchers acknowledged or incorporated lessons from prior waves of technological change in framing issues surrounding AI. As a proxy for this concept, we measured whether a given paper used the word "automation" in the abstract or title to capture whether the concept was important to the framing of the study. We found that most clusters did not mention the term at all in their abstracts or titles. The business-transformation cluster had the highest proportion of articles mentioning automation (19 %), followed by the work-impacts cluster (16 %). All other clusters were at 3 % or less. Overall, this indicator suggests that the organizational science literature can go further in explicitly placing AI into the context of longstanding trends in workplace automation.

Discussion of AI's effects on workers. Given our overall focus on AI and work, we also examined whether and how the literature discussed the potentially varied effects of AI on workers, given the established heterogeneity of implications it can transmit. We first examined whether researchers focused any of their analyses upon workers. Overall, most papers (67 %) did not explicitly or clearly address this topic. The variation within clusters is also informative. Papers in the decision-support and prediction-analysis clusters, which focused more on the development of the technology for specific applications, did not discuss the impacts of their focal AI upon workers (only 3–10 % of papers in these two clusters did so). Unsurprisingly, more than two-thirds of papers in the work-impacts cluster (71 %) discussed effects on workers, followed by the service-sector (55 %) and business-transformation (45 %) clusters.

For papers that did discuss worker impacts, we then investigated whether the focus was on positive, negative, or neutral effects, particularly given that such effects are often presented as conveying optimistic, pessimistic, or more balanced assessments of the technology itself. In terms of positive effects, several papers noted that AI provided enhanced support for decision makers ([69]; business-transformation cluster) and improved worker performance and wellbeing ([70]; business-transformation cluster). As for negative effects, papers referenced AI reducing employee engagement ([71]; work-impacts cluster), generating fear of job replacement ([72]; service-sector cluster) and job insecurity ([73]; business-transformation cluster), and leading to worker

exploitation ([74]; work-impacts cluster). Neutral effects generally focused on human–AI interaction and the overall benefits and costs associated with AI use (e.g., [75]; work-impacts cluster).

Of the 32 % of articles treating the issue of worker impacts, 34 % expressed positive impacts of AI, with 35 % presenting a negative view and the remainder conveying a neutral orientation (indicating that they discussed both benefits and drawbacks). Within clusters and among papers addressing the issue, the business-transformation and work-impacts clusters focused the most on the negative effects of AI on workers (48 % and 41 %, respectively). In terms of positive effects, papers in the service-sector and supply chain-operations clusters had the highest proportions (53 % and 46 %, respectively), while the work-impacts cluster had the lowest proportion (16 %; note that decision-support papers had no positive or negative worker impacts to report, reflecting our assessment that these clusters did not discuss these impacts). In the work-impacts and business-transformation clusters, the proportion of negative impact papers exceeded the positive impact proportion, while the reverse was true for clusters such as prediction-analysis and supply chain-operations. These outcomes align with our findings regarding the use of the substitution–complementarity–new task generation framework. That is, the work-impacts cluster shows signs of focusing more heavily upon technology's substitution (human replacing) effects than on any offsetting complementarity or new task generation effects.

To further understand how researchers characterize AI through its broader employment outcomes, we also examined whether papers in our sample discussed significant and widespread human unemployment as an outcome of AI.[9] Overall, only 6 % of articles explicitly discussed the idea that AI might lead to widespread technological unemployment. However, within the work-impacts and business-transformation clusters, 11–19 % contained language or conclusions consistent with this significant outcome. We further broadened our analysis of the characterization of AI's worker impacts to include either a prediction of negative worker impacts or a more general forecast of radical disruption, and the proportion of papers across the entire sample that contained such expectations rose to 28 %. In the work-impacts cluster, 60 % of articles conveyed these views, as opposed to 0 % of decision-support papers and 9 % of prediction-analysis studies.

Clarity indicators: Differences across empirical and non-empirical work. Because our manual coding captured paper methodologies, we also uncovered intersections between some of our *clarity* and *contextualization* indicators and the use of empirical methods. To be clear, this analysis is not asserting blanket value judgements regarding research approaches. Both empirical and non-empirical analyses in their many and varied forms provide important and indispensable contributions to scholarship. In the context of the study of AI and work, we aim only to report whether, within our sample, systematic differences were evident in the ways that empirical and non-empirical studies engaged with clarity indicators and to what outcomes (for methodological critiques in this area, see Deranty and Corbin, [29]).

Regarding general characterizations of AI, we found that characterizations such as "disruptive", "wholly new", "radically different", or "revolutionary" interacted with research methodology. Papers using empirical methods were 14 percentage points less likely than non-empirical articles to assert that AI is radically disruptive ($p < 0.01$). The work-impacts cluster was the least likely to use empirical methods, with just over half of papers employing these techniques (vs. over 90 % for the prediction-analysis cluster and over 70 % for the service-sector cluster). We do note, though, that some clusters are comparatively high in both empirical methods and assertions about disruptive AI. For example, in the business-transformation cluster, 83 % of papers use empirical methods, while 42 % use disruptive characterizations of AI. For the service-sector cluster, the equivalent percentages are 73 % (use of empirical methods) and 30 % (use of disruptive characterizations of AI). It is important to point out that, within these clusters, many papers show the same divergence that exists between clusters. For example, although precision is reduced due to the smaller sample size within clusters, in the business-transformation cluster, 55 % of non-empirical papers use disruptive characterizations, as opposed to 40 % of empirical papers. By the same token, 61 % of business-transformation papers that lack a concrete discussion of AI's capabilities have disruptive formulations, as opposed to 34 % of business-transformation papers that do contain such discussion. Regarding characterizations of AI's worker impacts, we analyzed whether papers included either a prediction of negative worker impacts or a more general forecast of radical disruption. Across our full sample, breaking down these papers by methodology, 43 % of the non-empirical papers report these significant negative conclusions, as opposed to 24 % of empirical studies.

2.3.2. *Reviewing clusters regarding contextualization of the phenomenon*

Focal levels of analysis. Our review identified the predominant level [s] of analysis employed in the study of AI to assess the degree of recognition that AI's effects will be contingent upon factors at multiple levels of analysis. We adopt levels of analysis definitions from the organizational science literature. For our purposes, micro-level research focuses on implications for the individual worker and/or work team and can include assessments of attitudes, behavior, and other psychological processes. Meso-level research is situated at the organizational level, including leadership and management, and can cover aspects such as strategy, firm capabilities, and business process efficiency. Macro-level research examines extra-organizational factors such as economy-related and industry-related aspects. Note that individual papers can address more than one level.

Beginning at the micro level, the work-impacts cluster focused most on this level (60 %), followed by the service-sector cluster (51 %), and the business-transformation cluster (42 %). The decision-support, prediction-analysis, and supply chain-operations clusters were the least likely to focus on AI's micro-level impacts (7–21 % of their papers did so). Around three-quarters of papers overall presented some discussion or analysis at the meso level, with the supply chain-operations and service-sector clusters showing the highest proportions (89 % and 85 %). The decision-support cluster was least likely to assess meso-level factors, although a majority of papers in this cluster nevertheless did so (62 %). While all clusters included some papers mentioning macro-level factors in their analyses, the proportion of such papers never rose above 50 %. The work-impacts cluster had the highest proportion of papers referencing macro-level factors (47 %), while the decision-support cluster had the lowest (18 %).

In examining the extent of multilevel research, or research focused on more than one level of analysis, we found that only 38 % of papers included two or more levels of analysis, and only 9 % of papers touched on all three levels. While the focus on a single level is arguably expected for a targeted cluster (e.g., supply chain-operations), this narrower framing is potentially more problematic for clusters that seek to assess AI, a phenomenon that cuts across multiple boundaries, and its impact on work. The work-impacts cluster did have the highest rate of multilevel analysis; however, one-third of the papers did not take multiple levels into account, and more than 80 % did not touch on the full range of levels.

Grounding in a specific industry or function. There were also differences among papers and clusters regarding the degree to which a given study was grounded in a particular industry or organizational function. Among papers in clusters such as prediction-analysis, service-sector, and supply chain-operations, more than 90 % of articles specifically delineated the function to which AI was being applied. In contrast, in the

[9] This coding question asked, "Does the paper indicate the authors think AI use will lead to technological unemployment/economy-wide job loss/widespread unemployment?" (Yes, No, Unclear). Within each paper the coding team searched for terms such as technological unemployment, economy-wide job loss, widespread unemployment.

work-impacts cluster the proportion that specified an AI function was 20 percentage points lower. Similarly, papers varied in the degree to which they situated their analyses of AI in an industry context. The service-sector, decision-support, supply chain-operations, and prediction-analysis clusters had the highest rates of precisely specified industries (66–75 %), while the work-impacts and business-transformation clusters had the lowest (52–58 %).

2.3.3. Distilling themes across clusters: application-orientation and generalized-orientation categories

This analysis of our manually coded sample ($n = 442$) offers evidence that our six clusters bifurcate based on several of our *clarity* and *contextualization* indicators, leading to differences in how they discuss and characterize AI and its implications for work. We suggest that this schism within organizational science research broadly reflects two categories of scholarship: *application-orientation* clusters and *generalized-orientation* clusters, where the bifurcation stems from their definitional and functional clarity regarding AI's capabilities and contextualizing their research within a specific function or industry.

The *application-orientation* group is constituted by the prediction-analysis, decision-support, supply chain-operations, and service-sector clusters. Compared with other clusters, these were the most likely to either define AI's capabilities concretely or situate their assessments within a specific function or industry. Generally, these clusters either focused on applying AI/neural network/machine learning applications to a specific (often predictive) problem, or situated their discussions of AI within a specific sector or function, such as in services or the supply chain function (see also [76] for the importance of such situated assessments). Compared with the other clusters, these clusters were less likely to characterize AI as a radically or wholly new and disruptive technology, less likely to contain claims regarding widespread technological unemployment resulting from AI, and less likely to focus on the negative (compared to the positive) outcomes of AI use for workers. At the same time, across all clusters, these were the least likely to engage in multilevel assessments of AI's effects and usually remained centered on a single level of analysis.

The *generalized-orientation* group consists of the work-impacts and business-transformation clusters. When compared across all clusters, these were the least likely to concretely define or discuss the capabilities of AI and the least likely to situate their assessments of AI's effects within a specific organizational function or industry. Compared with the *application-orientation* grouping, these clusters were also more likely to characterize AI as a radically or wholly new and disruptive technology, more likely to contain claims regarding widespread technological unemployment resulting from AI, and more likely to focus on the negative (compared to the positive) outcomes of AI use for workers. However, across all clusters, the work-impacts cluster was the most likely to engage in some type of multilevel analysis (although not across all three levels we assessed).

Our analyses also surfaced insights regarding how methodological choices across organizational science literature shape the assessed outcomes of AI. Specifically, compared with empirical papers, non-empirical work is more likely to focus on substitution processes (i.e., human replacement with AI) and to report negative or extreme effects for workers. These results suggest that, overall, the most dire assessments of AI's impact on work tended to come from non-empirical studies.

Taken together, these results go some way toward uncovering the technology characterizations, narratives, and approaches that different clusters of organizational science research are applying to the study of AI in relation to its worker and organizational impacts. The findings reinforce an ongoing challenge noted by Berente et al. [77], that technical and organizational fields tend to "black-box" each other's scholarship. We evaluate such challenges in the next section.

2.4. Evaluating the literature: issues springboarding the need for contextualized, multilevel assessments

Our review has presented some evidence on how the organizational science literature is approaching the study of AI at work. We now distill and discuss the ways in which the field's study of AI is constrained by issues relating to *clarifying* and *contextualizing* the phenomenon of AI at work.

2.4.1. Clarifying the phenomenon: defining and characterizing AI

Many articles in our review did not clearly define AI (i.e., what it is) nor concretely evaluate its functional capabilities (i.e., what it can do). While the lack of cohesive or consistently used definitions of AI is common [78], examining the impacts of a phenomenon requires its articulation [5]. Our identified groupings discussed above differ in their characterizations of AI. That is, the more a paper focused on the impact of AI on the future of work (i.e., the work-impacts cluster), the less likely that paper was to concretely define AI or discuss its technical capabilities, and the more likely it was to characterize AI as a radically new technology. Conversely, the more a paper discussed the technical development and capabilities of AI as applied to a specific problem (i.e., the prediction-analysis and decision-support clusters particularly), the less likely that paper was to discuss the broader implications of AI for work and the less likely it was to characterize AI as a radically new technology. For example, 42 % of work-impacts papers characterized AI as radically disruptive, while only 8 % of prediction-analysis papers did so (difference significant at the 99 % level). These results also reflect García-Peñalvo and Vázquez-Ingelmo's [79] finding that generative AI is commonly described in general terms by non-technical audiences, while the AI research community discusses specific generative models used. Such findings point to both the challenges generalists face in interpreting more technically oriented literature and the challenges technologists face in examining the wider implications of the technologies they design. These dynamics likely contribute to the lack of engagement across subdisciplines of the literature examining this phenomenon.

A question that then arises is why lack of clarification appears to be a common phenomenon. Although there are likely many reasons, given our focus on this issue we think it is important to posit specific ones around approaches to framing AI that appear to be muddying clarity. Further, to meet our call for reducing disciplinary silos and seeking technically informed capability discussions, we particularly draw on the computer science literature to highlight points around AI capabilities.

Contrasting broader definitions of AI with more technically focused ones surfaces one reason for potential mischaracterizations of AI's capabilities. Our suggestions here can help dig into the broader issues that Appio et al. [11] articulate regarding how ongoing dystopian narratives of AI can drive fear and anxiety through "misconceptions and exaggerated portrayals of AI's capabilities, as well as confusion about the degree of autonomy AI systems truly possess" (p. 4). When general definitions of AI in some way connect the phenomenon to mimicking, imitating, or even matching human thinking or intelligence, this could arguably feed concerns regarding human capabilities and wider existential risks [11]. While AIs can certainly do particular tasks better than humans, the way they do this through their current scope of capabilities is important to clarify.

Key components of AI's current technical capabilities and trajectory reflect an extension of past automation technologies with a strong focus on routinization and statistical pattern recognition [12,13]. This understanding was reflected more in our *application-orientation* clusters (e.g., [59], decision-support cluster) but was less evident in *generalized-orientation* clusters that make predictions about the future of work. It is important to note that this observation holds true even for more recent "reasoning" variations of large language models (large reasoning models, or LRMs). Research shows that these models show the statistical/probabilistic limitations of the LLMs from which they are derived

(evocatively referred to as the "embers of autoregression", [80], p. 1). While recent advances in generative AI, as well as "reasoning" models, have expanded expectations about the tasks that AI can perform, such advances can also generate a "good at language → good at thought" fallacy, conflating knowledge of the "statistical regularities of language" with "the ability to use language in real-world situations" ([81], p. 517). This itself can lead to misattributions of intelligence and intentionality to these systems [81].

Generative algorithms still do not "understand" the social or causal context in which they operate [82], and they remain dependent on large data-driven statistical correlations. As the computer scientist Jaron Lanier has noted, generative AI "can be thought of as illuminating previously hidden concordances between human creations, rather than as the invention of a new mind" [83]. Ultimately, though, this impressive modeling yields a data-intensive statistical engine that is vastly different from the human brain [84,85]. Again, this insight applies to the latest LRMs. Recent research from Apple also shows that even the latest LRMs still break down when complexity or context exceeds that of the training data [86]. Additional research demonstrates that LRM "performance improvements do not stem from the model learning general algorithmic procedures via demonstrations, but depend on carefully engineering highly problem specific prompts" ([87], p. 1). In the same vein, experimental research shows that intermediate "chain of thought" steps are not causally linked to model output [88]. As Kambhampati et al. [89, p. 9] note, the anthropomorphizing of intermediate-step displays in LRMs as "reasoning" or "thinking" obscures the underlying processes and "prevents researchers from understanding or improving how they actually work" and, taken further, their implications for human work.

Ultimately, characterizations of AI as reflecting human intelligence may then be driving concerns about AI's consumption of tasks that were previously the exclusive domain of human intelligence, another component of more dystopian AI narratives [11]. AI is effectively converting pieces of what may currently be non-routine processes into routine algorithms by using impressively sophisticated statistical correlations [90]. Its current uses essentially reflect that a wider range of work activities exhibit regular statistical patterns than was previously thought (see [16], p. 24) and so can have AI technologies applied to them. All this is not to say that some form of artificial general intelligence, a system "of equal intelligence to an adult human" ([91], p. 21), could not arise in the future, although this remains highly contested terrain [92]. However, for research focused on *present-day AI*, it is critical to rigorously assess current capabilities.

Recent AI developments certainly generate varied, wide-ranging, and significant implications for workers, and indeed raise important questions about the appropriate and ethical application of AI technologies to a widening range of work activities. This fact makes it particularly important to clarify the scope of AI's capabilities in order to set technically informed boundaries around its implications for human work (we canvas taxonomy options toward this end in Part 2 of the paper).

2.4.2. *Clarifying the phenomenon: historical context of technological change and work*

There is a long history of research on technological change and its effects on work that remains relevant for current studies of AI, and there is scope to employ such frameworks more consistently and in more balanced ways. For example, on our marker of engagement with the broad channels of replacement–complementarity–new task generation, in the work-impacts cluster half of papers mentioned substitution/replacement effects, while less than a quarter mentioned the ability of AI to generate new tasks. This finding mirrors a general trend toward focusing on the substitution effects of technology while comparatively neglecting the critical sister process by which AI stimulates demand for human labor in new functions (see [90]). This also aligns with wider work showing that binary utopian and dystopian AI narratives dominate public discourse, despite the value of more balanced narratives [11]. Disconnecting from past work on automation and computerization, which has consistently shown that complementarity and new task generation counterbalance substitution effects [17,93], may also be driven by the predominance of papers in the work-impacts cluster focusing on the novelty of AI. There is already commentary suggesting that generative AI's impacts may break away from these established models, such as by halving entry-level white-collar job opportunities and rocketing unemployment levels ([94], referencing the CEO of Anthropic, Dario Amodei), notwithstanding early evidence that its impacts are reflecting established patterns [95,96].

Overall, while there are clearly aspects of AI that differentiate it from past technologies and it is being applied to new areas of work not previously performed by technology, neglecting important lessons learned from how past waves of technological change have shaped work weakens the foundations from which to continue knowledge generation.

2.4.3. *Contextualizing the phenomenon: multilevel approaches to studying AI*

To fully understand AI's impacts on work requires incorporation of meso- and macro-level perspectives [25]. Such contingency-based approaches recognize that technology's effects are dependent upon a range of factors at different levels (e.g., [97]) and eschew technological determinism, or the view that technology's effects will always be felt in largely uniform ways. Contextualized research also provides a foundation to examine whether and how patterns of effects may exist across very different settings (see [98]). For example, anxiety about loss of livelihood (i.e., replacement) or control at work may be baseless in one industry or organization, while fully warranted in others, given the roles of economic forces and management implementation choices.

However, our analysis suggests little use of full multilevel research in the study of AI at work, with clusters' focal levels of analysis generally falling within research silos along established disciplinary lines. For example, examinations of AI's work impacts more frequently involved micro-level analysis, while the more industry-oriented clusters, such as supply chain-operations and service-sector, tended to focus on the meso level. However, single-level analyses do not fully illuminate nor connect the complex array of factors that generate a technology's workplace effects.

3. Part 2: groundwork for multilevel scholarship - illustrative, interdisciplinary, and multilevel framework for studying AI at work

Building from our critical assessment, we propose an illustrative, interdisciplinary, and multilevel framework (Fig. 3) that focuses on key factors that shape technology's work impacts. In discussing this framework, we suggest specific guideposts for future organizational AI scholarship that will contribute to a multilevel understanding of the phenomenon and strike a balance between *application-orientation* and *generalized-orientation* work. In essence, we seek to offer a pathway for creating another cluster of research: one that embraces interdisciplinarity and contextualized, contingent assessments of AI's workplace impacts.

We particularly build on the significant contemporary scholarship of Anthony et al. [[76], p. 1672], Bailey et al. [99], and Berente et al. [77] to ground our multilevel model in a "system" or "relational" view of AI's effects on work. A system view requires positioning AI "within a system of relations that extend outside of its immediate context" ([76], p. 1673). Like Anthony et al. [76], and reflecting the approach across our review work, our choice of factors at each level then centers on: (1) drawing out potential factors from relevant diverse disciplines, as examining the context-dependent impacts of new technologies means that "no one discipline … can capture the complexity" ([26], p. 340); and (2) drawing on established models of technology's effects to connect current AI research to the rich lineage of scholarship relevant to its contemporary study.

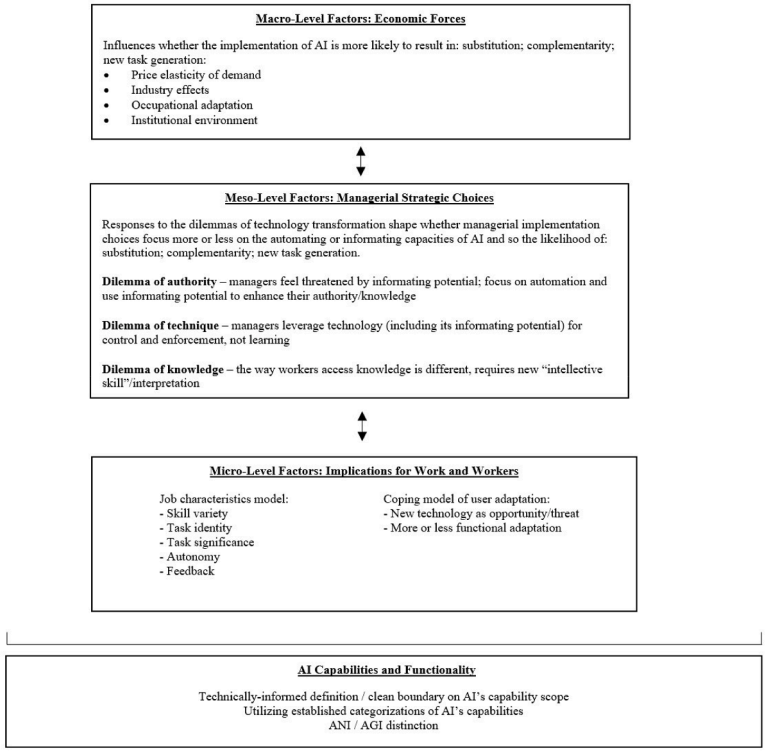

Fig. 3. Illustrative multilevel framework.

Adopting such an approach has challenges and limitations that are important to acknowledge. Undertaking interdisciplinary work is difficult, as it is often hampered by terminological differences, philosophical differences, the practical challenges of constructing interdisciplinary teams, and institutional systems that perpetuate disciplinary silos [100, 101]. Another limitation on constructing any multilevel model is that many factors are relevant to examining a phenomenon and no model is exhaustive. Using Subedi et al.'s [102] guidance, we explicitly acknowledge that as an interdisciplinary research team our choice of frameworks is guided by our expertise, and some may argue our disciplinary biases, but we aim to show how such choices can collectively go some way to generating richer insights.

As phenomenon-based research requires clarity on the focal phenomenon [5], our framework's foundation suggests pathways for clear characterizations of AI's capabilities to help *clarify* the phenomenon and its impacts. We particularly draw on information systems literature, as a key boundary-spanning discipline [77], to identify useful categorizations of AI's capabilities. We then integrate interdisciplinary perspectives on how technological processes affect work and workers across three levels of analysis to *contextualize* AI's study: economics-based models associated with capital/labor demand and investment patterns (macro level); social psychology and management research on managerial strategic choices for technology implementation (meso level); and job characteristics (organizational behavior) and processes of user adaptation to technology (information systems) that affect human workers' experiences of work (micro level). At each level of analysis, our spine remains the task-based model (substitution, complementarity, and new task generation; [17,68]) to reinforce how factors proposed at each level are relevant for, and connect to, the examination of AI's work impacts.

Overall, whether a researcher adopts a multilevel research design or simply acknowledges the limitations imposed by a particular macro-, meso-, or micro-setting, cognizance of multilevel factors is important for making sense of an overarching phenomenon such as AI.

3.1. *The phenomenon: the capabilities of AI and effects on work tasks*

Organizational science scholarship would benefit from "mov(ing) beyond symbolic narratives and abstract representations of AI" ([76], p. 1690). A key component of this is clarifying its capabilities and functionality in the context in which it is being used. Generalizations of AI can minimize its diverse applications and constrain precise examinations of its effects across such applications. Clarity can come from drawing on established categorizations of AI's capabilities that span broader to more fine-grained characterizations. Examples of the former include Kim et al. [103] delineating functional AI (i.e., AI that helps people complete particular tasks) and social AI (i.e., AI that acts as a social companion). Examples of the latter include Schmid et al.'s [104] characterization of types of AI across its methods (i.e., identifying simpler and more complex modeling and learning processes), capabilities (i.e., identifying what the AI is capable of doing, such as sensing, acting in some way, or communicating), and criticality (i.e., the level of risk in using AI methods/capabilities in particular ways). Function-specific characterizations also exist, such as Manis and Madhavaram [105] identifying how particular marketing activities can be enabled by AI functionality. More recently, Strobel et al. [106] suggest five distinct generative AI applications, from "generators" (which create content) and "reimaginators" (which reinterpret data to reflect a specific style), to "synthesizers" (which create synthetic data), and "assistants" (which provide domain-specific support). However, capabilities may not always be positive as, like any system, AI can fail. Raji et al.'s [107] failure taxonomy provides a counterpoint to the others by classifying the ways in which AI may cause harms, via poor execution across tasks, and engineering, deployment, and communication failures, including overstated capabilities. Such categorizations will need to adapt to AI's evolving performance and scope frontiers [77], but they provide a touchpoint for addressing clarification issues in the study of AI.

Clarity on its functionality then affords insights into AI's effects on human work through the substitution, complementarity, and new task generation framework. These channels are not mutually exclusive and

can co-occur. For example, an AI that automates (substitution) the screening of resumes could create more applicant insights for a recruiter (complementarity) and allow them to focus on more innovative selection techniques (new task generation). Connecting AI's capabilities to work tasks then sets a foundation for examining its impact on human work in a phenomenon-based way.

3.1.1. Future research guideposts: explicating the phenomenon

Scholars can adopt definitions that place clear boundaries around AI's technical capabilities and that differentiate it from human intelligence. While efforts at defining shifting technologies may appear futile, they offer the semblance of "ground truth" for their study. An example definition that could address the issues we discuss by providing research-relevant clarity could be the following: "Currently, AI is a technology that can remove the need for human intelligence, or complement it, in a specific task by reducing a problem, or part of it, to a pattern-based statistical algorithm." Such a formulation recognizes the source of AI's technical advantages while setting a clean boundary on its scope.

Researchers can also clearly articulate the type of AI being examined. This begins by differentiating between research that seeks to engage with speculative impacts regarding the potential of artificial general or super intelligence and research engaging with the concrete impacts of currently available forms of AI. Because both future-focused research on the possibilities of AI and examinations of current technologies already impacting businesses and society are critical and important areas of scholarship, it is necessary to clearly frame which area is the focus, without conflating advancements in narrow AI as the realization of artificial general intelligence, given their fundamental differences in capabilities and objectives. Articulating these distinctions can support definitional clarity of AI and help to ensure that studies of currently available forms of AI remain centered on its actual capabilities, while also allowing clearly future-oriented research to focus on the speculative potential of possible AI futures.

Additionally, researchers can leverage the types of categorization schemes identified above to clarify the functionality and application of AI in a given context to identify the types of tasks to which it is being applied. Ultimately, however, the degree, type, and location of substitution, complementarity, and new task generation will be heavily influenced by other multilevel factors, and we first turn to those at the macro level.

3.2. Macro-level factors: demand, industry, occupational, and institutional effects

Where our reviewed papers referenced macro-level factors this often involved broad discussions, such as how technology advancements may influence the economy [73] and the future of work [108]. Fully addressing contextualization issues necessitates targeted attention to the macro-level context in which AI is implemented. We cannot cover every macro-level factor that is relevant for a given piece of research, but we can give illustrative examples of the types of demand, occupational, industry, and institutional effects that are critical to consider.

One of the most important and overlooked macro factors is the price elasticity of demand for a given product or service [109,110]. The impact of AI-driven automation on employment and job structure within a given industry will vary considerably based on whether consumers will demand more of the industry's product or service when the price falls due to productivity gains. For a product where consumer demand is less sensitive to price (e.g., milk production), automation will result in substantial within-industry job loss and substitution effects will likely dominate. In contrast, for a product where demand is very sensitive to price changes (e.g., game consoles or sports car production), automation will result in increased sales volume and industry expansion. Some routine jobs may still be lost in these latter industries due to substitution effects, but historically the enhanced product demand in such cases will heighten use of the complementarity and new task generation channels (and hence employment growth overall). Taking these factors into account is very relevant, even for a researcher studying a micro-oriented topic such as worker anxiety due to technological change. In the above-mentioned examples, worker fears of job loss are more likely to be justified in the inelastic case, while there would be a higher probability in the elastic case that worker anxiety reflects a general fear of change rather than a realistic appraisal of industry fortunes.

Even in the case of an industry with price-inelastic products, the impact of AI may well increase aggregate employment outside of the industry in question. AI-driven cost reductions increase both profits and, to the extent that these profits are spent or invested, demand for other products and their associated jobs [90]. Thus, in assessing AI's impact on work, it is important to take macro-level elasticities into account and to specify whether the analysis is narrowly limited to within-industry effects or inclusive of broad cross-industry effects.

Beyond demand elasticities and industry effects, attending to occupational adaptation is important. Where AI use may diminish employment in some jobs that are worker- and worksite-specific, the occupation is not necessarily doomed. Almost all jobs consist of a variety of tasks [111], and this task composition may change. Most occupations survive over time by focusing on complementary tasks or new functions, while shedding routine characteristics [112]. The degree to which an occupation adapts to changing technology is influenced by the nature of the technology (clarity aspect) as well as macro-level factors such as industry growth prospects, worker bargaining power, and the flexibility of educational institutions.

Finally, the institutional environment is critical. The deployment of seemingly "neutral" technologies varies widely based on the nature of a country's or region's institutions. Existing research has documented that usage of once-cutting-edge technologies such as magnetic resonance imaging (MRI) machines differs greatly across countries based on the structure and incentives of national health systems [113]. Acemoglu and Restrepo [114] show negative employment effects from robot adoption in the United States, while Adachi et al. [115] demonstrate positive effects in Japan. Recent survey evidence indicates that AI adoption attitudes and investment vary substantially by country as well [116]. Educational institutions, the level of unionization, laws, regulations, and a host of other institutional factors will affect AI's impacts at the macro, meso, and micro levels. Building on the example of micro-focused worker anxiety research above, the nature and extent of technologically-driven anxiety will likely differ in an environment in which unions are present to bargain for retention and retraining. Referencing such factors will support results that are more robust to changes in context.

Our discussion of macro factors is not meant to be exhaustive, with additional country effects and the business cycle being examples of other relevant factors (e.g., [110,117]). We aim to demonstrate that demand elasticities, industry scope, occupational trends, and institutional configurations, among other macro factors, provide important context and interaction effects for AI research at any level of analysis.

3.2.1. Future research guideposts: connecting macro to meso/micro levels of analysis

Although meso- and micro-level research can occur within a given industry setting, it is rare to see the impact of macro factors that vary across industries explicitly incorporated into an examination of the more micro impacts of AI on work. In its fullest embrace, this would mean scholars utilizing research models that incorporate product-market demand factors, industry characteristics, and occupational trends into analyses that may be primarily targeted at the micro or meso levels. This could mean choosing research sites based upon industry characteristics that reflect not only the likelihood of human task replacement, but also complementarity and new task generation. In addition, this will help rebalance research toward examining the effects of different task channels while acknowledging the degree to which sectoral contexts

make such channels more or less likely. In its narrowest embrace, explicitly recognizing that demand, industry, and occupational trends will influence the use of AI in a given context will acknowledge these as important boundary conditions.

3.3. Meso-level factors: managerial choices and dilemmas of technological transformation

Technology is implemented within a relational context, in which a phenomenon's effects depend upon the connections within which it is embedded [99]. In organizations (and beyond), this means that issues surrounding power, politics, and status influence technology use [76, 118]. Indeed, Berente et al. [77] highlight the facets of AI (autonomy, learning, and inscrutability) that are heightening the challenges of organizing.

Such issues are not new, and here we draw on Zuboff's [97] "two faces" of intelligent technologies that generate dilemmas for organizational knowledge, authority, and power structures that ultimately influence its use at work. While other scholars' meso-level insights are certainly also relevant here, we focus on Zuboff's work to emphasize that appropriate clarification of the phenomenon involves recognition of the historical and intellectual lineage of that phenomenon. She posits that, in their *automating* capacity, the effects of intelligent technologies are little different from those felt with the earliest factory-based machines; they "replace the human body with a technology that enables the same processes to be performed with more continuity and control" ([97], p. 9), akin to substitution. The same technology can also *informate*, generating more information about a work process, increasing transparency of activities, creating new forms and quality of data, and generating new innovations [97], akin to worker complementarity and new task generation. Automating and informating capacities are not mutually exclusive, as technology used for informating often builds on its use for automation.

Technology transformations result in three dilemmas [97] that are, in part, resolved through managerial strategic choices to adopt either the automating or informating powers of a technology [4,23,119]. These decisions will then influence how technologies such as AI are deployed toward task effect channels. The *dilemma of knowledge* reflects that technology changes how knowledge is developed, received, interpreted, and shared. For example, in white collar work, technology can replace or complement existing forms of knowledge captured physically (e.g., paper-based sources) or implicitly (e.g., human intuition). To effectively utilize these changed forms of knowledge through informating requires organizations to actively develop workers' "intellective skills," which organizations may or may not do well ([97], p. 16). In the latter case, workers may proactively learn about the system to develop their skills, including doing so outside of a manager's oversight ([97]; see also [120]).

The informating potential of intelligent technologies can then be perceived as opportunities for learning, innovation, and improvement [121] or as threats to managerial power, creating a *dilemma of authority*. As workers' intellective skills improve, work processes become more transparent, and as knowledge becomes more accessible, managers can perceive threats to their traditional authority and power [97], leading them to focus more on the technology's automating capacities [121] to retain control over information and re-affirm existing power structures. The *dilemma of technique* stems from these challenges to authority, in that managers use technology's informating potential not to improve work processes, but to leverage the increased visibility of work and worker actions afforded to heighten control and surveillance [97]. The reciprocal nature of responses to these two dilemmas may lead employees to respond to the technology defensively, by manipulating or subverting its use, usually creating suboptimal outcomes overall [121].

The choices managers make in addressing these dilemmas should influence how task effect channels are utilized. For example, if an AI is implemented with a focus on its automating capacities it will tend to degrade job quality by overemphasizing substitution effects (also bearing in mind macro-level factors). In contrast, AI implementation with a focus on its informating capacities creates opportunities to enhance workers' jobs through access to new forms of information and new tasks – via complementarity and new task generation – thereby affirming the value human workers continue to bring to work processes. Overall, Zuboff's [97] work demonstrates just one example of a rich prior framework applicable to examining how strategic choices shape AI's effects.

3.3.1. Future research guideposts: connecting meso to micro levels of analysis

Several features of AI will impact the nature of Zuboff's [97] dilemmas and their resolution and so offer opportunities for future research. In terms of *knowledge*, AI advances reflect the new and increasingly sophisticated informating opportunities these technologies bring. Better connecting research on specific AI applications with work that assesses the context in which technology is deployed will help more fully assess how AI can promote an enriched and informated work environment and how this translates for workers through complementarity and new task generation channels. Conversely, the inscrutability of AI [77] may constrain the new and complementary forms of knowledge AI can generate and how effectively workers can interpret them, making examinations of human–AI knowledge co-development important [23].

Changes in knowledge also precipitate changes in *skills*. Zuboff [[97], p. 101] differentiates workers' "acting-on" skills, associated with explicit and direct action upon physical materials, equipment, or data, and "acting-with" skills, associated with softer skills of communication, negotiation, and interaction. With AI use come ethical and fairness concerns associated with "datafication of the workplace" ([42], p. 2), whereby knowledge and acting-on skills that deal with quantifiable metrics and abstract information become dominant and so diminish the use of more implicit knowledge associated with acting-with skills [97]. Such concerns are heightened when the "efficiency driven logic" and predictive capabilities of AI (e.g., when used for predicting machine maintenance) are extended to managing people (e.g., predicting "good recruits," "good leaders," and "good" worker behavior; [122], p. 1; [42, 123]), where such assessments arguably require acting-with skills to promote fairness perceptions [124,125]. Examining the use of AI in specific roles, and how managerial choices lead to it substituting or complementing particular skill sets, will facilitate assessments of how changes in knowledge affect work and either enhance, diminish, or require new human skills to do it.

3.4. Micro-level factors: effects of technology on work and workers

Many micro-level theories exist that can help examine the worker impacts of technology. We do not, and could not, canvas that scope here. What the above contextualization can help support, however, is targeted application of different disciplinary micro-level theories. For example, historical and contemporary work highlight the various and competing effects of technology upon workers, such as how it can enhance or diminish autonomy [18,97] and lessen or intensify work demands [126, 127]. These experiences can be broadly captured under, for example, the job characteristics model (JCM), a seminal framework identifying five core job characteristics that generate worker motivation, satisfaction, and performance [128]. Specifically, skill variety and use (i.e., the amount and type of skills activated in work), task identity (i.e., the number of integrated tasks completed), and task significance (i.e., the perceived impact of work on others) all contribute to experiencing *meaningful work*. Autonomy (i.e., levels of independence and empowerment) heightens experiences of *responsibility* and, finally, feedback (i.e., knowledge of one's performance at work) heightens *one's knowledge of the job*.

While the immediate effects of AI use on job design and job quality

can be captured through models such as the JCM, past research shows that users also engage in adaptation strategies during technology implementation. At the micro level, this also makes it important to position workers as agentic in their use, or otherwise, of a new technology. For example, Beaudry and Pinsonneault's [129] coping model of user adaptation outlines how end users first appraise a new technology as an opportunity or a threat and then engage in ways to adapt to its use either more or less functionally, ultimately leading to outcomes such as effective use through to organizational exit. While not exhaustive, utilizing such established models helps to build upon existing knowledge and avoid neglecting technological continuity.

3.4.1. Future research guideposts: connecting macro and meso to micro levels of analysis

Overall, clarifying the nature of a phenomenon such as AI allows us to leverage insights from models that were designed to evaluate prior technological changes in the workplace. Given the broad and complex origins of AI, micro factors will also interact with other levels of analysis. Our work also suggests how AI, through task effect channels, can lead to different job quality and adaptation outcomes. For example, in a sector with inelastic product demand and managerial choices based on an automating focus (i.e., substitution), there is a greater chance that job quality will be degraded (potentially including job loss) and workers may disengage from adaptation strategies. In sectors with elastic product demand and managerial choices based on leveraging AI's informating capacities by enhancing workers' capabilities (emphasizing complementarity and new task generation), there is an increased likelihood that job characteristics such as skill variety and use, autonomy, and task significance will be enhanced and workers' cognitive and behavioral adaptation will lead to more effective use of the technology. Research designs that incorporate these multilevel perspectives have the potential to offer further insights into why the same AI exerts particular effects in some workplaces, but different ones elsewhere.

4. Limitations

Beyond the boundaries of our analyses already noted, our work is focused on disciplines within the organizational science literature, meaning we did not engage with the widest possible scope of disciplinary contributions to the topic (see Deranty and Corbin, [29] for such efforts). Our scope was also limited to assessments of *clarity* and *contextualization* and specific markers within those attributes, and we acknowledge that other attributes exist and could be canvassed in future work. Like any study of technology, our examination reflects literature and technological capabilities up to a point in time (for us, the start of 2024). As the frontiers of those technological capabilities change, this will bring new questions and potentially new narratives within specific disciplines that only ongoing work in this area will uncover. Finally, our multilevel model does not, and cannot, articulate every aspect relevant to our topic, but aims to further prompt and widen contextualized thinking in this area.

5. Conclusions

The impact of AI on the future of work is a topic of significant interest and concern for management, organizations, and employees. Our review offers some evidence of how organizational science literature is conceptualizing, characterizing, and investigating AI at work to better understand and evaluate how this scholarship *clarifies* and *contextualizes* the phenomenon. Notably, we find that research clusters within what we label *application-orientation* and *generalized-orientation* categories. The former grouping was more likely to clarify and contextualize its research within a specific function or industry and was less likely to make dire predictions about AI's effect on work. The latter grouping was less likely to reference AI's capabilities or situate their analyses in a specific industry context and was more likely to contain dire assessments of AI's impact on work. We argue that it is essential to clarify AI's capabilities and expand scholarship toward multilevel research designs that situate it within a broader context that recognizes macro, meso, and micro factors, while acknowledging its continuity with prior literature on technological change, automation, and work. Phenomenon-based, interdisciplinary, and multilevel approaches are vital for developing a comprehensive understanding of AI's effects. Bridging levels of analysis will also help inform various stakeholders, such as policymakers, leaders, and technology designers, about the multifaceted impacts that their technologies, policies, and strategies have on workers. Such research is now critical to provide a balanced assessment of an important but often misunderstood technological phenomenon.

CRediT authorship contribution statement

Sarah Bankins: Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Andrew Weaver:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Mauricio Marrone:** Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Simon Lloyd D. Restubog:** Writing – review & editing, Project administration, Methodology, Formal analysis. **Sang Eun Woo:** Writing – review & editing, Formal analysis.

Declaration of competing interest

The authors have no competing nor financial interests to declare.

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Appendix 1. Distribution of Literature Sources by Journal (Publications with $n > 5$)

Journal Title	Number of Publications	% of Total Sample
Technological Forecasting and Social Change	81	9 %
International Journal of Production Research	75	8 %
Journal of Business Research	63	7 %
Decision Support Systems	51	5 %
International Journal of Production Economics	35	4 %
Industrial Marketing Management	27	3 %
Production Planning and Control	24	3 %
Manufacturing and Service Operations Management	22	2 %
Production and Operations Management	21	2 %

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Journal Title	Number of Publications	% of Total Sample
International Journal of Contemporary Hospitality Management	18	2 %
MIS Quarterly: Management Information Systems	16	2 %
International Journal of Hospitality Management	16	2 %
Strategic Management Journal	15	2 %
Human Resource Management Review	13	1 %
Management Science	13	1 %
Research Policy	13	1 %
New Technology, Work and Employment	13	1 %
Information and Management	12	1 %
Journal of the Operational Research Society	12	1 %
IEEE Transactions on Engineering Management	11	1 %
Information Systems Research	11	1 %
Tourism Management	11	1 %
Leadership Quarterly	11	1 %
International Journal of Human Resource Management	10	1 %
Technovation	10	1 %
Journal of Management Information Systems	10	1 %
Journal of Business Ethics	9	1 %
Journal of Service Research	9	1 %
Personnel Psychology	8	1 %
Journal of Information Technology	8	1 %
British Journal of Management	8	1 %
Psychology and Marketing	7	1 %
International Journal of Forecasting	7	1 %
European Journal of Marketing	7	1 %
Journal of Operations Management	7	1 %
Organizational Research Methods	7	1 %
Journal of Product Innovation Management	7	1 %
Energy Economics	6	1 %
Transportation Research Part E: Logistics and Transportation Review	6	1 %
Decision Sciences	6	1 %
Journal of Strategic Information Systems	6	1 %
Others	213	23 %

Appendix 2. Publications Ranked by Annual Citation Rate Within Research Clusters (2021–2023)

Cluster 1 – Prediction-analysis

Authors	Title	Year	Journal
Choi T.-M.; Kumar S.; Yue X.; Chan H.-L.	Disruptive Technologies and Operations Management in the Industry 4.0 Era and Beyond	2022	*Production and Operations Management*
Haefner N.; Wincent J.; Parida V.; Gassmann O.	Artificial intelligence and innovation management: A review, framework, and research agenda	2021	*Technological Forecasting and Social Change*
Choudhury P.; Foroughi C.; Larson B.	Work-from-anywhere: The productivity effects of geographic flexibility	2021	*Strategic Management Journal*
Li K.; Mai F.; Shen R.; Yan X.	Measuring Corporate Culture Using Machine Learning	2021	*Review of Financial Studies*
Sautner Z.; Van Lent L.; Vilkov G.; Zhang R.	Firm-Level Climate Change Exposure	2023	*Journal of Finance*

Cluster 2 - Business-transformation

Authors	Title	Year	Journal
Saura J.R.; Palacios-Marques D.; Ribeiro-Soriano D.	Exploring the boundaries of open innovation: Evidence from social media mining	2023	*Technovation*
Bag S.; Pretorius J.H.C.; Gupta S.; Dwivedi Y.K.	Role of institutional pressures and resources in the adoption of big data analytics powered artificial intelligence, sustainable manufacturing practices and circular economy capabilities	2021	*Technological Forecasting and Social Change*
Kim S.S.; Kim J.; Badu-Baiden F.; Giroux M.; Choi Y.	Preference for robot service or human service in hotels? Impacts of the COVID-19 pandemic	2021	*International Journal of Hospitality Management*
Chatterjee S.; Rana N.P.; Dwivedi Y. K.; Baabdullah A.M.	Understanding AI adoption in manufacturing and production firms using an integrated TAM-TOE model	2021	*Technological Forecasting and Social Change*
Kamble S.S.; Gunasekaran A.; Kumar V.; Belhadi A.; Foropon C.	A machine learning based approach for predicting blockchain adoption in supply Chain	2021	*Technological Forecasting and Social Change*

Cluster 3 – Service-sector

Authors	Title	Year	Journal
Puntoni S.; Reczek R.W.; Giesler M.; Botti S.	Consumers and Artificial Intelligence: An Experiential Perspective	2021	*Journal of Marketing*
Chowdhury S.; Dey P.; Joel-Edgar S.; Bhattacharya S.; Rodriguez-Espindola O.; Abadie A.; Truong L.	Unlocking the value of artificial intelligence in human resource management through AI capability framework	2023	*Human Resource Management Review*
Pereira V.; Hadjielias E.; Christofi M.; Vrontis D.	A systematic literature review on the impact of artificial intelligence on workplace outcomes: A multi-process perspective	2023	*Human Resource Management Review*
Rana N.P.; Chatterjee S.; Dwivedi Y.K.; Akter S.	Understanding dark side of artificial intelligence (AI) integrated business analytics: assessing firm's operational inefficiency and competitiveness	2022	*European Journal of Information Systems*
Mustak M.; Salminen J.; Ple L.; Wirtz J.	Artificial intelligence in marketing: Topic modeling, scientometric analysis, and research agenda	2021	*Journal of Business Research*

Cluster 4 – Work-impacts

Authors	Title	Year	Journal
Raisch S.; Krakowski S.	Artificial intelligence and management: The automation-augmentation paradox	2021	*Academy of Management Review*
Vrontis D.; Christofi M.; Pereira V.; Tarba S.; Makrides A.; Trichina E.	Artificial intelligence, robotics, advanced technologies and human resource management: a systematic review	2022	*International Journal of Human Resource Management*
Huang M.-H.; Rust R.T.	Engaged to a Robot? The Role of AI in Service	2021	*Journal of Service Research*
Margherita A.	Human resources analytics: A systematization of research topics and directions for future research	2022	*Human Resource Management Review*
Krakowski S.; Luger J.; Raisch S.	Artificial intelligence and the changing sources of competitive advantage	2023	*Strategic Management Journal*

Cluster 5 - Supply chain-operations

Authors	Title	Year	Journal
Ivanov D.; Tang C.S.; Dolgui A.; Battini D.; Das A.	Researchers' perspectives on Industry 4.0: multi-disciplinary analysis and opportunities for operations management	2021	*International Journal of Production Research*
Hu H.; Xu J.; Liu M.; Lim M.K.	Vaccine supply chain management: An intelligent system utilizing blockchain, IoT and machine learning	2023	*Journal of Business Research*
Helo P.; Hao Y.	Artificial intelligence in operations management and supply chain management: an exploratory case study	2022	*Production Planning and Control*
Dubey R.; Bryde D.J.; Foropon C.; Tiwari M.; Dwivedi Y.; Schiffling S.	An investigation of information alignment and collaboration as complements to supply chain agility in humanitarian supply chain	2021	*International Journal of Production Research*
Belhadi A.; Kamble S.; Fosso Wamba S.; Queiroz M.M.	Building supply-chain resilience: an artificial intelligence-based technique and decision-making framework	2022	*International Journal of Production Research*

Cluster 6 – Decision-support

Authors	Title	Year	Journal
Toorajipour R.; Sohrabpour V.; Nazarpour A.; Oghazi P.; Fischl M.	Artificial intelligence in supply chain management: A systematic literature review	2021	*Journal of Business Research*
Park J.; Chun J.; Kim S.H.; Kim Y.; Park J.	Learning to schedule job-shop problems: representation and policy learning using graph neural network and reinforcement learning	2021	*International Journal of Production Research*
Rolf B.; Jackson I.; Muller M.; Lang S.; Reggelin T.; Ivanov D.	A review on reinforcement learning algorithms and applications in supply chain management	2023	*International Journal of Production Research*
Kazancoglu Y.; Sagnak M.; Mangla S.K.; Sezer M.D.; Pala M.O.	A fuzzy based hybrid decision framework to circularity in dairy supply chains through big data solutions	2021	*Technological Forecasting and Social Change*
Mintz O.; Gilbride T.J.; Lenk P.; Currim I.S.	The right metrics for marketing-mix decisions	2021	*International Journal of Research in Marketing*

References

[1] N. Maslej, L. Fattorini, R. Perrault, V. Parli, A. Reuel, E. Brynjolfsson, J. Etchemendy, K. Liggett, T. Lyons, J. Manyika, J.C. Niebles, Y. Shoham, R. Wald, J. Clark, Artificial intelligence index report 2024 (2024), https://aiindex.stanford.edu/wp-content/uploads/2024/05/HAI_AI-Index-Report-2024.pdf.

[2] A. Hanelt, R. Bohnsack, D. Marz, C. Antunes Marante, A systematic review of the literature on digital transformation: insights and implications for strategy and organizational change, J. Manag. Stud. 58 (5) (2021) 1159–1197, https://doi.org/10.1111/joms.12639.

[3] B.J. Copeland. Artificial intelligence, Encyclopedia Britannica, 2020.

[4] M.H. Jarrahi, Artificial intelligence and the future of work: human-AI symbiosis in organizational decision making, Bus. Horiz. 61 (4) (2018) 577–586, https://doi.org/10.1016/j.bushor.2018.03.007.

[5] G. von Krogh, Artificial intelligence in organizations: new opportunities for phenomenon-based theorizing, Acad. Manag. Discov. 4 (4) (2018) 404–409, https://doi.org/10.5465/amd.2018.0084.

[6] E. Broad, We need to let go of regulating 'artificial intelligence', The Innovation Papers, (August 2024), Retrieved from: https://www.innovationaus.com/we-need-to-let-go-of-regulating-artificial-intelligence/.

[7] S. Cave, C. Craig, K. Dihal, S. Dillon, J. Montgomery, B. Singler, L. Taylor, Portrayals and Perceptions of AI and Why They Matter, The Royal Society, 2018. https://royalsociety.org/~/media/policy/projects/ai-narratives/AI-narratives-workshop-findings.pdf.

[8] V. Tangermann, Apple Researchers Just Released a Damning Paper That Pours Cold Water on the Entire AI Industry, Futurism, 2025. Retrieved from: https://futurism.com/apple-damning-paper-ai-reasoning.
[9] H. Al Haddi, AI washing: the cultural traps that lead to exaggeration and how CEOs can stop them, Calif. Manag. Rev. Insights (2024). Retrieved from: https://cmr.berkeley.edu/2024/12/ai-washing-the-cultural-traps-that-lead-to-exaggeration-and-how-ceos-can-stop-them/.
[10] S. Khan, Research report EPR. AI in Wonderland: Engineering in the Age of Overpromised Technology 2025002, 2025. 10.4271/EPR2025002.
[11] F.P. Appio, C.T. Hernandez, F. Platania, F. Schiavone, AI narratives in fiction and media: exploring thematic parallels in public discourse, Technovation 142 (2025) 103201, https://doi.org/10.1016/j.technovation.2025.103201.
[12] A. Darwiche, *arXiv (Cornell University)*. Human-Level Intelligence or Animal-Like Abilities?, 2017. 10.48550/arxiv.1707.04327.
[13] G. Marcus, E. Davis, Rebooting AI: Building Artificial Intelligence We Can Trust, 1st edition, Pantheon Books, 2019.
[14] J. Clifton, A. Glasmeier, M. Gray, When machines think for us: the consequences for work and place, Camb. J. Regions Econ. Soc. 13 (1) (2020) 3–23, https://doi.org/10.1093/cjres/rsaa004.
[15] S. Feuerriegel, J. Hartmann, C. Janiesch, P. Zschech, Generative AI, Bus. Inf. Syst. Eng. 66 (1) (2024) 111–126, https://doi.org/10.1007/s12599-023-00834-7.
[16] E. Mollick, Co-Intelligence: Living and Working With AI, Penguin Random House, NY, 2024.
[17] D. Acemoglu, P. Restrepo, Automation and new tasks: how technology displaces and reinstates labor, J. Econ. Perspect. 33 (2) (2019) 3–30, https://doi.org/10.1257/jep.33.2.3.
[18] K.C. Kellogg, M.A. Valentine, A. Christin, Algorithms at work: the new contested terrain of control, Acad. Manag. Ann. 14 (1) (2020) 366–410, https://doi.org/10.5465/annals.2018.0174.
[19] R.F. Zammuto, T.L. Griffith, A. Majchrzak, D.J. Dougherty, S. Faraj, Information technology and the changing fabric of organization, Organ. Sci. 18 (5) (2007) 749–762, https://doi.org/10.1287/orsc.1070.0307.
[20] S.S. Poloz, Technological progress and monetary policy: managing the fourth industrial revolution, J. Int. Money. Finance 114 (2021) 102373, https://doi.org/10.1016/j.jimonfin.2021.102373.
[21] S. Bankins, P. Formosa, The ethical implications of artificial intelligence (AI) for meaningful work, J. Bus. Ethics 185 (2023) 725–740, https://doi.org/10.1007/s10551-023-05339-7.
[22] G. Bell, The metaverse is a new word for an old idea, MIT Technol. Rev. (2022). https://www.proquest.com/other-sources/metaverse-is-new-word-old-idea/docview/2820961446/se-2?accountid=12219.
[23] M.H. Jarrahi, In the age of the smart artificial intelligence: AI's dual capacities for automating and informating work, Bus. Inf. Rev. 36 (4) (2019) 178–187, https://doi.org/10.1177/0266382119883999.
[24] J.C. Molloy, R.E. Ployhart, P.M. Wright, The myth of "the" micro-macro divide: bridging system-level and disciplinary divides, J. Manage 37 (2) (2011) 581–609, https://doi.org/10.1177/0149206310365000.
[25] S. Raisch, S. Krakowski, Artificial intelligence and management: the automation–augmentation paradox, Acad. Manag. Rev. 46 (1) (2021) 192–210, https://doi.org/10.5465/AMR.2018.0072.
[26] A. Rip, R. Kemp, Technological change. Human Choice and Climate Change: Vol. II, Resources and Technology, Battelle Press, 1998, pp. 327–399.
[27] S. Faraj, P.M. Leonardi, Strategic organization in the digital age: rethinking the concept of technology, Strateg. Organ. 20 (4) (2022) 771–785, https://doi.org/10.1177/14761270221130253.
[28] Y.L. Bavik, B. Shao, A. Newman, G. Schwarz, Crisis leadership: a review and future research agenda, Leadersh. Q. 32 (6) (2021) 101518, https://doi.org/10.1016/j.leaqua.2021.101518.
[29] J.P. Deranty, T. Corbin, Artificial intelligence and work: a critical review of recent research from the social sciences, AI Soci. 39 (2024) 675–691, https://doi.org/10.1007/s00146-022-01496-x.
[30] M.R.W. Hiebl, Sample selection in systematic literature reviews of management research, Organ. Res. Methods 26 (2) (2023) 229–261, https://doi.org/10.1177/1094428120986851.
[31] N.J. van Eck, L. Waltman, CitNetExplorer: a new software tool for analyzing and visualizing citation networks, J. Informetr. 8 (4) (2014) 802–823, https://doi.org/10.1016/j.joi.2014.07.006.
[32] A. Kumar, R.D. Gopal, R. Shankar, K.H. Tan, Fraudulent review detection model focusing on emotional expressions and explicit aspects: investigating the potential of feature engineering, Decis. Support. Syst. 155 (2022) 113728, https://doi.org/10.1016/j.dss.2021.113728.
[33] M. Stevenson, C. Bravo, Advanced turbidity prediction for operational water supply planning, Decis. Support. Syst. 119 (2019) 72–84, https://doi.org/10.1016/j.dss.2019.02.009.
[34] R.M. van Steenbergen, M.R.K. Mes, Forecasting demand profiles of new products, Decis. Support. Syst. 139 (2020) 113401, https://doi.org/10.1016/j.dss.2020.113401.
[35] S. Chatterjee, R. Chaudhuri, D. Vrontis, A. Thrassou, S.K. Ghosh, Adoption of artificial intelligence-integrated CRM systems in agile organizations in India, Technol. Forecast. Soc. Change 168 (2021) 120783, https://doi.org/10.1016/j.techfore.2021.120783.
[36] M.S. Talukder, G. Sorwar, Y. Bao, J.U. Ahmed, M.A.S. Palash, Predicting antecedents of wearable healthcare technology acceptance by elderly: a combined SEM-neural network approach, Technol. Forecast. Soc. Change 150 (2020) 119793, https://doi.org/10.1016/j.techfore.2019.119793.
[37] C. Trocin, I.V. Hovland, P. Mikalef, C. Dremel, How artificial intelligence affords digital innovation: a cross-case analysis of Scandinavian companies, Technol. Forecast. Soc. Change 173 (2021) 121081, https://doi.org/10.1016/j.techfore.2021.121081.
[38] P. Mikalef, K. Conboy, J. Krogstie, Artificial intelligence as an enabler of B2B marketing: a dynamic capabilities micro-foundations approach, Ind. Mark. Manag. 98 (2021) 80–92, https://doi.org/10.1016/j.indmarman.2021.08.003.
[39] C. Wilson-Nash, A. Goode, A. Currie, Introducing the socialbot: a novel touchpoint along the young adult customer journey, Eur. J. Mark. 54 (10) (2020) 2621–2643, https://doi.org/10.1108/EJM-07-2019-0555.
[40] D. Yin, M. Li, H. Qiu, Do customers exhibit engagement behaviors in AI environments? The role of psychological benefits and technology readiness, Tour. Manage 97 (2023) 104745, https://doi.org/10.1016/j.tourman.2023.104745.
[41] I. Aleksander, Partners of humans: a realistic assessment of the role of robots in the foreseeable future, J. Inf. Technol. 32 (1) (2017) 1–9, https://doi.org/10.1057/s41265-016-0032-4.
[42] U. Gal, T.B. Jensen, M.-K. Stein, Breaking the vicious cycle of algorithmic management: a virtue ethics approach to people analytics, Inf. Organ. 30 (2) (2020) 100301, https://doi.org/10.1016/j.infoandorg.2020.100301.
[43] A.J. Wood, M. Graham, V. Lehdonvirta, I. Hjorth, Good gig, bad gig: autonomy and algorithmic control in the global gig economy, Work Employ. Soc. 33 (1) (2019) 56–75, https://doi.org/10.1177/0950017018785616.
[44] O. Allal-Chérif, V. Simón-Moya, A.C.C. Ballester, Intelligent purchasing: how artificial intelligence can redefine the purchasing function, J. Bus. Res. 124 (2021) 69–76, https://doi.org/10.1016/j.jbusres.2020.11.050.
[45] W. Fang, Y. Guo, W. Liao, K. Ramani, S. Huang, Big data driven jobs remaining time prediction in discrete manufacturing system: a deep learning-based approach, Int. J. Prod. Res. 58 (9) (2020) 2751–2766, https://doi.org/10.1080/00207543.2019.1602744.
[46] Y. Jiang, T. Feng, Y. Huang, Antecedent configurations toward supply chain resilience: the joint impact of supply chain integration and big data analytics capability, J. Oper. Manag. 70 (2) (2024) 257–284, https://doi.org/10.1002/joom.1282.
[47] A. Bansal, R.J. Kauffman, R.R. Weitz, Comparing the modeling performance of regression and neural networks as data quality varies: a business value approach, J. Manag. Inf. Syst. 10 (1) (1993) 11–32, https://doi.org/10.1080/07421222.1993.11517988.
[48] T. Lawrence, Impacts of artificial intelligence on organizational decision making, J. Behav. Decis. Mak. 4 (3) (1991) 195–214, https://doi.org/10.1002/bdm.3960040306.
[49] T.M. Willems, L.E.M.W. Brandts, Sub-symbolic management decision support system: an application of neural networks in management, Prod. Plan. Control 8 (2) (1997) 123–132, https://doi.org/10.1080/095372897235389.
[50] K. Kim, O.J. Park, S. Yun, H. Yun, What makes tourists feel negatively about tourism destinations? Application of hybrid text mining methodology to smart destination management, Technol. Forecast. Soc. Change 123 (2017) 362–369, https://doi.org/10.1016/j.techfore.2017.01.001.
[51] W. Leal Filho, T. Wall, S.A.R. Mucova, G.J. Nagy, A.L. Balogun, J.M. Luetz, A. W. Ng, M. Kovaleva, F. Mohammad Safiul Azam, F. Alves, Z. Guevara, N. R. Matandirotya, A. Skouloudis, A. Tzachor, K. Malakar, O. Gandhi, Deploying artificial intelligence for climate change adaptation, Technol. Forecast. Soc. Change 180 (2022) 121662, https://doi.org/10.1016/j.techfore.2022.121662.
[52] P. Gaczek, G. Leszczyński, A. Mouakher, Collaboration with machines in B2B marketing: overcoming managers' aversion to AI-CRM with explainability, Ind. Mark. Manag. 115 (2023) 127–142, https://doi.org/10.1016/j.indmarman.2023.09.007.
[53] M.M. Mariani, M. Borghi, Artificial intelligence in service industries: customers' assessment of service production and resilient service operations, Int. J. Prod. Res. 62 (15) (2023) 5400–5416, https://doi.org/10.1080/00207543.2022.2160027.
[54] F. Bonetti, M. Montecchi, K. Plangger, H.J. Schau, Practice co-evolution: collaboratively embedding artificial intelligence in retail practices, J. Acad. Mark. Sci. 51 (4) (2023) 867–888, https://doi.org/10.1007/s11747-022-00896-1.
[55] U. Pamuksuz, J.T. Yun, A. Humphreys, A brand-new look at you: predicting brand personality in social media networks with machine learning, J. Interact. Mark. 56 (1) (2021) 1–15, https://doi.org/10.1016/j.intmar.2021.05.001.
[56] A.L. Fayard, Notes on the meaning of work: labor, work, and action in the 21st century, J. Manage Inq. 30 (2) (2021) 207–220, https://doi.org/10.1177/1056492619841705.
[57] S. Shady, T. Kaihara, N. Fujii, D. Kokuryo, A novel feature selection for evolving compact dispatching rules using genetic programming for dynamic job shop scheduling, Int. J. Prod. Res. 60 (13) (2022) 4025–4048, https://doi.org/10.1080/00207543.2022.2053603.
[58] J. Bode, Decision support with neural networks in the management of research and development: concepts and application to cost estimation, Inf. Manag. 34 (1) (1998) 33–40, https://doi.org/10.1016/S0378-7206(98)00043-3.
[59] W.Y.K. Chiang, D. Zhang, L. Zhou, Predicting and explaining patronage behavior toward web and traditional stores using neural networks: a comparative analysis with logistic regression, Decis. Support. Syst. 41 (2) (2006) 514–531, https://doi.org/10.1016/j.dss.2004.08.016.
[60] S.W. Palocsay, M.M. White, Neural network modeling in cross-cultural research: a comparison with multiple regression, Organ. Res. Methods 7 (4) (2004) 389–399, https://doi.org/10.1177/1094428104268030.
[61] L. Pettersen, Why artificial intelligence will not outsmart complex knowledge work, Work Employ. Soc. 33 (6) (2019) 1058–1067, https://doi.org/10.1177/0950017018817489.

[62] M.P. Durango-Gutiérrez, J. Lara-Rubio, A. Navarro-Galera, Analysis of default risk in microfinance institutions under the Basel III framework, Int. J. Finance Econ. 28 (2) (2023) 1261–1278, https://doi.org/10.1002/ijfe.2475.

[63] P. Dizikes, What Do We Know About the Economics of AI? MIT News, 2024. Retrieved from: https://news.mit.edu/2024/what-do-we-know-about-economics-ai-1206.

[64] B. Marr, The AI Revolution: How Predictive, Prescriptive, and Generative AI Are Reshaping Our World, a, Forbes, 2024. Retrieved from: https://www.forbes.com/sites/bernardmarr/2024/10/15/the-ai-revolution-how-predictive-prescriptive-and-generative-ai-are-reshaping-our-world/.

[65] B. Eastwood, Sam Altman believes AI will change the world (and everything else). Ideas made to matter (May 2024), Retrieved from: https://mitsloan.mit.edu/ideas-made-to-matter/sam-altman-believes-ai-will-change-world-and-everything-else.

[66] A. Agrawal, J. Gans, A. Goldfarb, Generative AI is still just a prediction machine, Harv. Bus. Rev. (2024). Retrieved from: https://hbr.org/2024/11/generative-ai-is-still-just-a-prediction-machine.

[67] C. Last, Global commons in the global brain, Technol. Forecast. Soc. Change 114 (2017) 48–64, https://doi.org/10.1016/j.techfore.2016.06.013.

[68] D.H. Autor, F. Levy, R.J. Murnane, The skill content of recent technological change: an empirical exploration, Q. J. Econ. 118 (4) (2003) 1279–1333, https://doi.org/10.1162/003355303322552801.

[69] T. Grønsund, M. Aanestad, Augmenting the algorithm: emerging human-in-the-loop work configurations, J. Strateg. Inf. Syst. 29 (2) (2020) 101614, https://doi.org/10.1016/j.jsis.2020.101614.

[70] S.S. Kamble, A. Gunasekaran, A. Ghadge, R. Raut, A performance measurement system for industry 4.0 enabled smart manufacturing system in SMMEs: a review and empirical investigation, Int. J. Prod. Econ. 229 (2020) 107853, https://doi.org/10.1016/j.ijpe.2020.107853.

[71] A. Braganza, W. Chen, A. Canhoto, S. Sap, Productive employment and decent work: the impact of AI adoption on psychological contracts, job engagement and employee trust, J. Bus. Res. 131 (2021) 485–494, https://doi.org/10.1016/j.jbusres.2020.08.018.

[72] G. Rampersad, Robot will take your job: innovation for an era of artificial intelligence, J. Bus. Res. 116 (2020) 68–74, https://doi.org/10.1016/j.jbusres.2020.05.019.

[73] T. Nam, Technology usage, expected job sustainability, and perceived job insecurity, Technol. Forecast. Soc. Change 138 (2019) 155–165, https://doi.org/10.1016/j.techfore.2018.08.017.

[74] T. Vieira, Platform couriers' self-exploitation: the case study of Glovo, New. Technol. Work Employ. 38 (3) (2023) 493–512, https://doi.org/10.1111/ntwe.12272.

[75] Y.R. Shrestha, V. Krishna, G. von Krogh, Augmenting organizational decision-making with deep learning algorithms: principles, promises, and challenges, J. Bus. Res. 123 (2021) 588–603, https://doi.org/10.1016/j.jbusres.2020.09.068.

[76] C. Anthony, B.A. Bechky, A.L. Fayard, "Collaborating" with AI: taking a system view to explore the future of work, Organ. Sci. 34 (5) (2023) 1672–1694, https://doi.org/10.1287/orsc.2022.1651.

[77] N. Berente, B. Gu, J. Recker, R. Santhanam, Managing artificial intelligence, MIS Q. 45 (3) (2021), https://doi.org/10.25300/MISQ/2021/16274.

[78] C. Collins, D. Dennehy, K. Conboy, P. Mikalef, Artificial intelligence in information systems research: a systematic literature review and research agenda, Int. J. Inf. Manage 60 (2021) 102383, https://doi.org/10.1016/j.ijinfomgt.2021.102383.

[79] F.J. García-Peñalvo, A. Vázquez-Ingelmo, What do we mean by GenAi? A systematic mapping of the evolution, trends, and techniques involved in generative AI, Int. J. Interact. Multimed. Artif. Intell. 8 (4) (2023) 7–16, https://doi.org/10.9781/ijimai.2023.07.006.

[80] R.T. McCoy, S. Yao, D. Friedman, M.D. Hardy, T.L. Griffiths, Embers of autoregression show how large language models are shaped by the problem they are trained to solve, Proce. Nat. Acad. Sci. 121 (41) (2014) e2322420121, https://doi.org/10.1073/pnas.2322420121.

[81] K. Mahowald, A.A. Ivanova, I.A. Blank, N. Kanwisher, J.B. Tenenbaum, E. Fedorenko, Dissociating language and thought in large language models, Trends Cogn. Sci. (Regul. Ed.) (2024), https://doi.org/10.1016/j.tics.2024.01.011.

[82] A. Chemero, LLMs differ from human cognition because they are not embodied, Nat. Hum. Behav. 7 (11) (2023) 1828–1829, https://doi.org/10.1038/s41562-023-01723-5.

[83] J. Lanier. There is no A.I, New Yorker, 2023.

[84] J. Pearl, D. Mackenzie, The Book of Why: The New Science of Cause and Effect, Firest edition, Basic Books, 2018.

[85] B.C. Smith, The Promise of Artificial intelligence: Reckoning and Judgment, 1 ed., MIT Press, 2019 https://doi.org/10.7551/mitpress/12385.001.0001.

[86] P. Shojaee, I. Mirzadeh, K. Alizadeh, M. Horton, S. Bengio, M. Farajtabar, The illusion of thinking: understanding the strengths and limitations of reasoning models via the lens of problem complexity (2025), *arXiv preprint arXiv: 2506.06941*.

[87] K. Stechly, K. Valmeekam, S. Kambhampati, Chain of thoughtlessness? An analysis of CoT in planning, in: Proceedings of the *Thirty-Eighth Annual Conference on Neural Information Processing Systems*, 2025.

[88] G. Bao, H. Zhang, L. Yang, C. Wang, Y. Zhang, LLMs with chain-of-thought are non-causal reasoners. arXiv e-prints, arXiv-2402 (2024), Retrieved from: https://arxiv.org/html/2402.16048v1.

[89] S., Kambhampati, K. Stechly, K. Valmeekam, L. Saldyt, S. Bhambri, V. Palod, A. Gundawar, S. Samineni, D. Kalwar, U. Biswas, Stop anthropomorphizing intermediate tokens as reasoning/thinking traces! arXiv preprint arXiv: 2504.09762 (2025), 10.48550/arXiv.2504.09762.

[90] D.H. Autor, Why are there still so many jobs? The history and future of workplace automation, J. Econ. Perspect. 29 (3) (2015) 3–30, https://doi.org/10.1257/jep.29.3.3.

[91] T. Walsh, N. Levy, G. Bell, A. Elliott, J. Maclaurin, I. Mareels, F. Wood, The effective and ethical development of artificial intelligence (2019). https://acola.org/wp-content/uploads/2019/07/hs4_artificial-intelligence-report.pdf.

[92] B. Marr, The Important Difference Between Generative AI and AGI, Forbes, 2024. Retrieved from: https://www.forbes.com/sites/bernardmarr/2024/05/08/the-important-difference-between-generative-ai-and-agi/.

[93] K. Hötte, M. Somers, A. Theodorakopoulos, Technology and jobs: a systematic literature review, Technol. Forecast. Soc. Change 194 (2023) 122750, https://doi.org/10.1016/j.techfore.2023.122750.

[94] B. Montgomery, Will AI Wipe Out the First Rung of the Career ladder? The Guardian, 2025. Retrieved from: https://www.theguardian.com/global/2025/jun/02/artificial-intelligence-jobs-techscape.

[95] Hartley, J., Jolevski, F., Melo, V., and Moore, B. (2025). The labor market effects of generative artificial intelligence. Available at SSRN: https://ssrn.com/abstract=5136877 or http://dx.doi.org/10.2139/ssrn.5136877.

[96] N. Salari, M. Beiromvand, A. Hosseinian-Far, J. Habibi, F. Babajani, M. Mohammadi, Impacts of generative artificial intelligence on the future of labor market: a systematic review, Comput. Hum. Behav. Rep. (2025) 100652, https://doi.org/10.1016/j.chbr.2025.100652.

[97] S. Zuboff, In the Age of the Smart Machine: The Future of Work and Power, Basic Books, 1988.

[98] M. Beane, The Skill Code: How to Save Human Ability in An Age of Intelligent Machines, HarperCollins, USA, 2024.

[99] D.E. Bailey, S. Faraj, P.J. Hinds, P.M. Leonardi, G. von Krogh, We are all theorists of technology now: a relational perspective on emerging technology and organizing, Organ. Sci. 33 (1) (2022) 1–18, https://doi.org/10.1287/orsc.2021.1562.

[100] K.L. Daniel, M. McConnell, A. Schuchardt, M.E. Peffer, Challenges facing interdisciplinary researchers: findings from a professional development workshop, PLoS One 17 (4) (2022) e0267234, https://doi.org/10.1371/journal.pone.0267234.

[101] L. Eisenberg, T.C. Pellmar (Eds.), Bridging Disciplines in the Brain, Behavioral, and Clinical Sciences, National Academy Press, Washington, D.C, 2000.

[102] J. Subedi, J.B. Houston, K. Sherman-Morris, Interdisciplinary research as an iterative process to build disaster systems knowledge, Risk Anal. 41 (7) (2021) 1072–1077, https://doi.org/10.1111/risa.13244.

[103] J. Kim, K. Merrill Jr, C. Collins, AI as a friend or assistant: the mediating role of perceived usefulness in social AI vs. functional AI, Telemat. Inform. 64 (2021) 101694, https://doi.org/10.1016/j.tele.2021.101694.

[104] T. Schmid, W. Hildesheim, T. Holoyad, K. Schumacher, The AI methods, capabilities and criticality grid: a three-dimensional classification scheme for artificial intelligence applications, KI. Künstl. Intell. (Oldenbourg) 35 (3–4) (2021) 425–440, https://doi.org/10.1007/s13218-021-00736-4.

[105] K.T. Manis, S. Madhavaram, AI-Enabled marketing capabilities and the hierarchy of capabilities: conceptualization, proposition development, and research avenues, J. Bus. Res. 157 (2023) 113485, https://doi.org/10.1016/j.jbusres.2022.113485.

[106] G. Strobel, L. Banh, F. Möller, T. Schoormann, Exploring generative artificial intelligence: a taxonomy and types, in: Proceedings of the *Hawaii International Conference on System Sciences 2024 (HICSS 2024)*, Hawaii, USA, 2024.

[107] I.D. Raji, I.E. Kumar, A. Horowitz, A. Selbst, The fallacy of AI functionality, in: Proceedings of the ACM Conference on Fairness, Accountability, and Transparency, New York, NY, USA, 2022.

[108] D. Minbaeva, Disrupted HR? Hum. Resour. Manag. Rev. 31 (4) (2021) 100820 https://doi.org/10.1016/j.hrmr.2020.100820.

[109] J. Bessen, AI and jobs: the role of demand, in: NBER Working Paper Series, 24235, National Bureau of Economic Research, 2018.

[110] E. Brynjolfsson, T. Mitchell, What can machine learning do? Workforce implications: profound change is coming, but roles for humans remain, Science (1979) 358 (6370) (2017) 1530–1534, https://doi.org/10.1126/science.aap8062.

[111] M. Arntz, T. Gregory, U. Zierahn, The risk of automation for jobs in OECD countries: a comparative analysis. OECD Social, Employment and Migration Working Papers, no.189, OECD Publishing, 2016.

[112] J. Bessen, Learning By Doing: The Real Connection Between Innovation, Wages, and Wealth, Yale University Press, 2015.

[113] G.F. Anderson, P. Hussey, V. Petrosyan, It's still the prices, stupid: why the US spends so much on health care, and a tribute to UWE Reinhardt, Health Aff. 38 (1) (2019) 87–95, https://doi.org/10.1377/hlthaff.2018.05144.

[114] D. Acemoglu, P. Restrepo, Robots and jobs: evidence from US labor markets, J. Political Econ. 128 (6) (2020) 2188–2244, https://doi.org/10.1086/705716, arXiv:2506.06941.

[115] D. Adachi, D. Kawaguchi, Y.U. Saito, Robots and employment: evidence from Japan, 1978–2017, J. Labor. Econ. 42 (2) (2024) 591–634, https://doi.org/10.1086/723205.

[116] J. Loucks, S. Hupfer, D. Jarvis, T. Murphy, Future in the Balance? How Countries Are Pursuing An AI Advantage, Deloitte Insights, 2019. https://www2.deloitte.com/us/en/insights/focus/cognitive-technologies/ai-investment-by-country.html.

[117] C. Lloyd, J. Payne, Rethinking country effects: robotics, AI and work futures in Norway and the UK, New Technol. Work Employ. 34 (3) (2019) 208–225, https://doi.org/10.1111/ntwe.12149.
[118] D.E. Bailey, S.R. Barley, Beyond design and use: how scholars should study intelligent technologies, Inf. Organ. 30 (2) (2020) 100286, https://doi.org/10.1016/j.infoandorg.2019.100286.
[119] W.J. Orlikowski, The duality of technology: rethinking the concept of technology in organizations, Organ. Sci. 3 (3) (1992) 398–427, https://doi.org/10.1287/orsc.3.3.398.
[120] M. Beane, Shadow learning: building robotic surgical skill when approved means fail, Adm. Sci. Q. 64 (1) (2018) 87–123, https://doi.org/10.1177/0001839217751692.
[121] A. Burton-Jones, What have we learned from the smart machine? Inf. Organ. 24 (2) (2014) 71–105, https://doi.org/10.1016/j.infoandorg.2014.03.001.
[122] L.M. Giermindl, F. Strich, O. Christ, U. Leicht-Deobald, A. Redzepi, The dark sides of people analytics: reviewing the perils for organisations and employees, Eur. J. Inf. Syst. 31 (3) (2021) 410–435, https://doi.org/10.1080/0960085X.2021.1927213.
[123] A. Lee, I. Inceoglu, O. Hauser, M. Greene, Determining causal relationships in leadership research using machine learning: the powerful synergy of experiments and data science, Leadersh. Q. 33 (5) (2022) 101426, https://doi.org/10.1016/j.leaqua.2020.101426.
[124] S. Bankins, P. Formosa, Y. Griep, D. Richards, AI decision making with dignity? Contrasting workers' justice perceptions of human and AI decision making in a human resource management context, Inf. Syst. Front. 24 (3) (2022) 857–875, https://doi.org/10.1007/s10796-021-10223-8.
[125] P. Tambe, P. Cappelli, V. Yakubovich, Artificial intelligence in human resources management: challenges and a path forward, Calif. Manage Rev. 61 (4) (2019) 15–42, https://doi.org/10.1177/0008125619867910.
[126] B.C. Amick, O. Ostberg, Office automation, occupational stress and health: a literature analysis with specific attention to expert systems, Inf. Technol. People 3 (3) (1987) 191–209, https://doi.org/10.1108/eb022648.
[127] S.K. Parker, G. Grote, Automation, algorithms, and beyond: why work design matters more than ever in a digital world, Appl. Psychol. 71 (4) (2022) 1171–1204, https://doi.org/10.1111/apps.12241.
[128] J.R. Hackman, G.R. Oldham, Motivation through the design of work: test of a theory, Organ. Behav. Hum. Perform 16 (2) (1976) 250–279, https://doi.org/10.1016/0030-5073(76)90016-7.
[129] A. Beaudry, A. Pinsonneault, Understanding user responses to information technology: a coping model of user adaptation, MIS Q. 29 (3) (2005) 493–524, https://doi.org/10.2307/25148693.

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